Experiment No: 1

Name: Navya Nair **Roll No. : 24**

Batch: B **Performance Date : 20/07/26**
Submission Date:2/08/26

Topic:	Installing Hadoop in Pseudo Distributed Mode & getting familiar with Hadoop HDFS commands.
Prerequisite:	Basic knowledge of Hadoop is required.
Mapping With COs:	CSL7012.1
Objective:	• To acquire the knowledge of different components present in Hadoop Ecosystem. • To understand the Hadoop 2.x architecture. • To learn how to set up and configure Hadoop in Pseudo Distributed • Mode. • To learn how to work with the Hadoop HDFS file system. • Getting familiar with Hadoop HDFS commands.
Outcome:	• Students will be able to differentiate between the local file system and HDFS. • Students will be able to install Hadoop clusters and configure it in any installation mode. • Students will be able to start and work with the Hadoop HDFS file system using various Hadoop commands.
Instructions:	This experiment is a compulsory experiment. All the students are required to perform this experiment individually.
Deliverables:	**Submission on Moodle:** 1. Explain all Hadoop installation modes. Hadoop can be installed and run in three different modes: **1. Standalone (Local) Mode** • It is the default mode of Hadoop and does not require any configuration. • Hadoop runs as a single Java process and uses the **local file system** instead of HDFS. • It is mainly used for learning, testing, and debugging MapReduce applications on a single machine. **2. Pseudo Distributed Mode** • In this mode, Hadoop runs on a single machine, but each Hadoop service (such as NameNode, DataNode, ResourceManager, and NodeManager) runs as a separate Java process. • It uses HDFS instead of the local file system, simulating a multi-node Hadoop cluster on one computer. • This mode is commonly used for development, testing, and educational purposes before deploying a real cluster.

3. Fully Distributed Mode

- In this mode, Hadoop is installed on multiple machines (a cluster) connected through a network.
- One machine acts as the master node (running services like NameNode and ResourceManager), while the other machines act as worker nodes (running DataNode and NodeManager services).
- It provides high performance, fault tolerance, scalability, and distributed storage and processing, making it suitable for production environments and large-scale Big Data applications.

2. List down components of Hadoop Cluster.

The main components of a Hadoop cluster are:

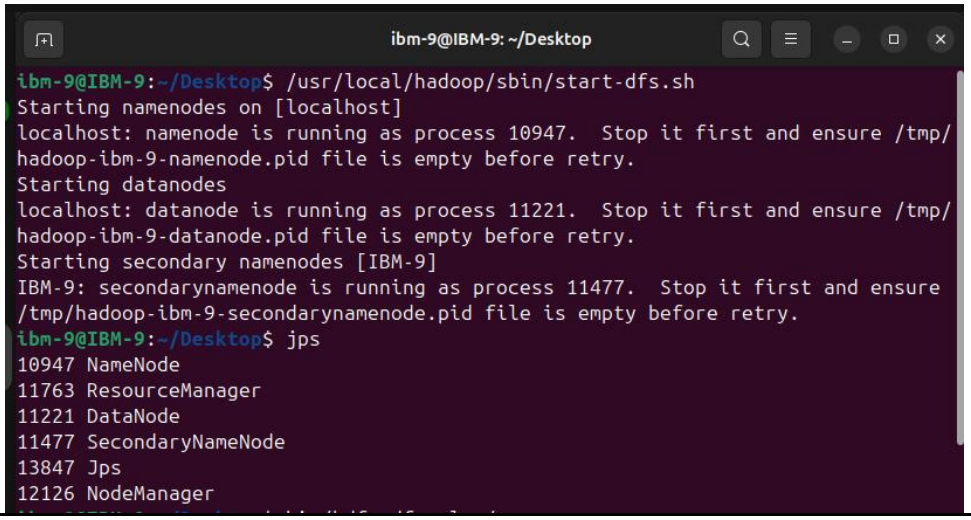
1. **NameNode**
    - The master node of HDFS.
    - Manages the file system namespace and stores metadata about files and directories.
    - Keeps track of where data blocks are stored in the cluster.
2. **DataNode**
    - The worker node of HDFS.
    - Stores the actual data blocks.
    - Performs read and write operations as instructed by the NameNode.
3. **Secondary NameNode**
    - Creates periodic checkpoints of the NameNode metadata.
    - Helps reduce the size of the edit logs and assists in recovery.
    - It is **not** a backup of the NameNode.
4. **ResourceManager**
    - The master component of YARN (Yet Another Resource Negotiator).
    - Manages cluster resources and schedules applications across the cluster.
5. **NodeManager**
    - Runs on every worker node.
    - Monitors resource usage (CPU, memory, etc.) and executes application containers assigned by the ResourceManager.
6. **HDFS (Hadoop Distributed File System)**

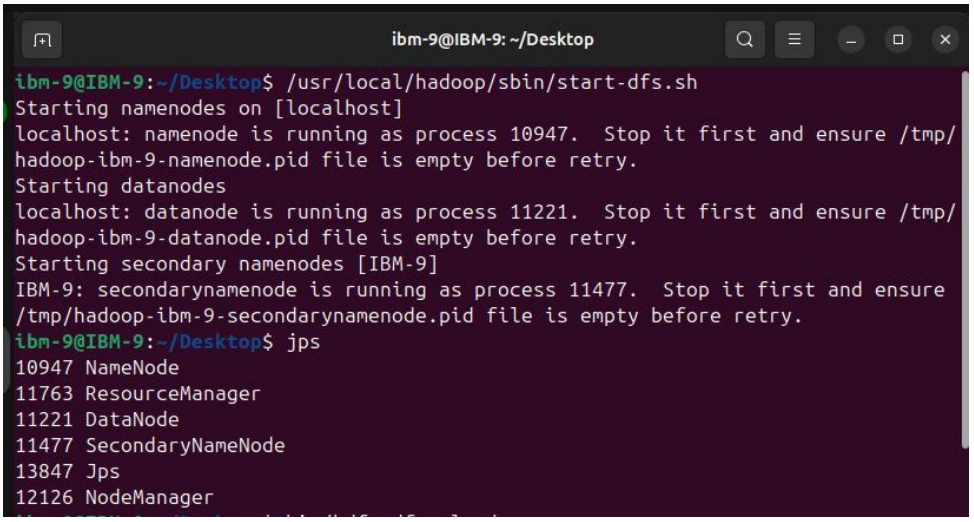
- Hadoop's distributed storage system.
- Stores large datasets across multiple machines with data replication for fault tolerance.

7. **YARN (Yet Another Resource Negotiator)**
   - Handles resource management and job scheduling.
   - Allocates resources to applications and manages their execution across the cluster.
8. **MapReduce**
   - Hadoop's data processing framework.
   - Processes large datasets in parallel using the Map and Reduce phases.

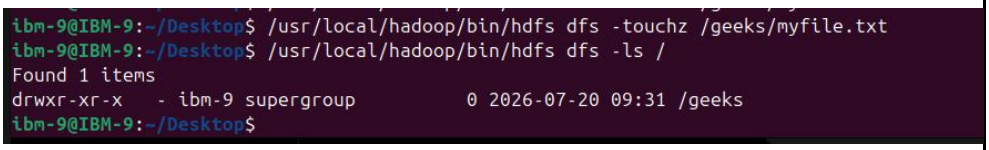
3. Take a snapshot of each step of hadoop installation and for all HDFS commands with input and its output.

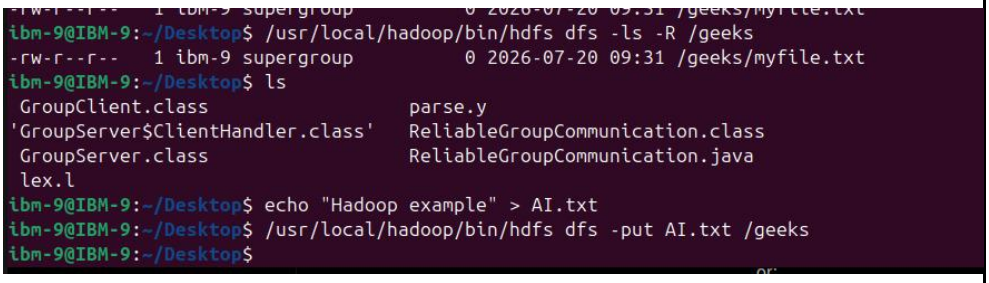
Step 1: Check HDFS Directory Permissions

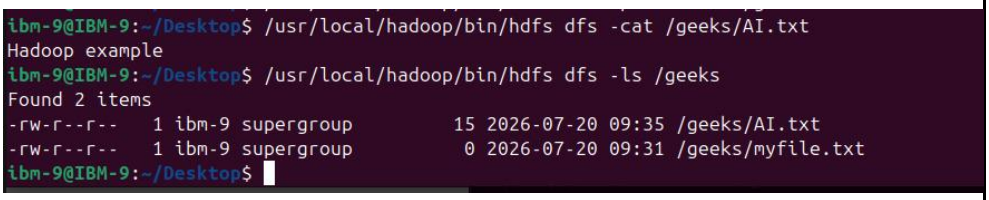


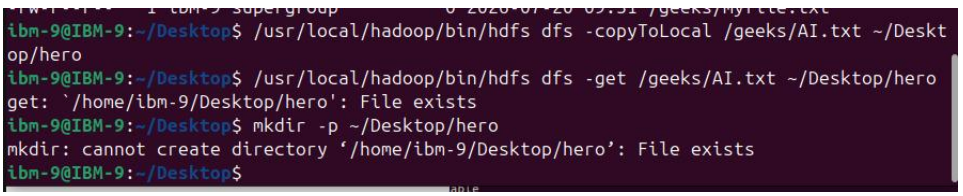

ibm-9@IBM-9: ~/Desktop
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/sbin/start-dfs.sh
Starting namenodes on [localhost]
localhost: namenode is running as process 10947. Stop it first and ensure /tmp/hadoop-ibm-9-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 11221. Stop it first and ensure /tmp/hadoop-ibm-9-datanode.pid file is empty before retry.
Starting secondary namenodes [IBM-9]
IBM-9: secondarynamenode is running as process 11477. Stop it first and ensure /tmp/hadoop-ibm-9-secondarynamenode.pid file is empty before retry.
ibm-9@IBM-9:~/Desktop$ jps
10947 NameNode
11763 ResourceManager
11221 DataNode
11477 SecondaryNameNode
13847 Jps
12126 NodeManager

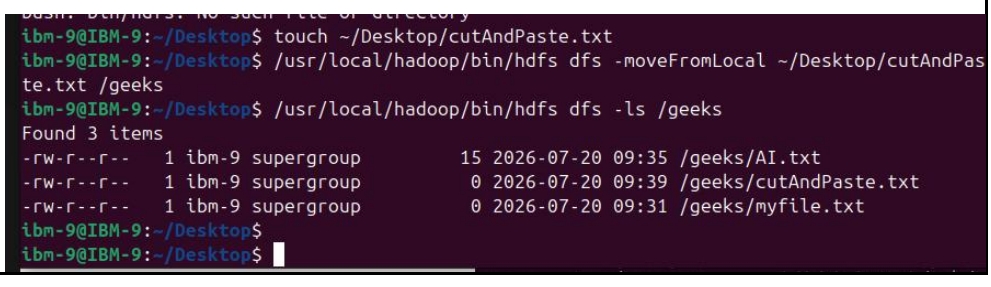

ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -ls /
ibm-9@IBM-9:~/Desktop$ jps
10947 NameNode
11763 ResourceManager
11221 DataNode
11477 SecondaryNameNode
14123 Jps
12126 NodeManager


ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -touchz /geeks/myfile.txt
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -ls /
Found 1 items
drwxr-xr-x - ibm-9 supergroup 0 2026-07-20 09:31 /geeks
ibm-9@IBM-9:~/Desktop$


ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -ls -R /geeks
-rw-r--r-- 1 ibm-9 supergroup 0 2026-07-20 09:31 /geeks/myfile.txt
ibm-9@IBM-9:~/Desktop$ ls
GroupClient.class parse.y
'GroupServer$ClientHandler.class' ReliableGroupCommunication.class
GroupServer.class ReliableGroupCommunication.java
lex.l
ibm-9@IBM-9:~/Desktop$ echo "Hadoop example" > AI.txt
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -put AI.txt /geeks
ibm-9@IBM-9:~/Desktop$


ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -cat /geeks/AI.txt
Hadoop example
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -ls /geeks
Found 2 items
-rw-r--r-- 1 ibm-9 supergroup 15 2026-07-20 09:35 /geeks/AI.txt
-rw-r--r-- 1 ibm-9 supergroup 0 2026-07-20 09:31 /geeks/myfile.txt
ibm-9@IBM-9:~/Desktop$


ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -copyToLocal /geeks/AI.txt ~/Desktop/hero
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -get /geeks/AI.txt ~/Desktop/hero
get: `/home/ibm-9/Desktop/hero': File exists
ibm-9@IBM-9:~/Desktop$ mkdir -p ~/Desktop/hero
mkdir: cannot create directory ‘/home/ibm-9/Desktop/hero’: File exists
ibm-9@IBM-9:~/Desktop$


ibm-9@IBM-9:~/Desktop$ touch ~/Desktop/cutAndPaste.txt
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -moveFromLocal ~/Desktop/cutAndPaste.txt /geeks
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -ls /geeks
Found 3 items
-rw-r--r-- 1 ibm-9 supergroup 15 2026-07-20 09:35 /geeks/AI.txt
-rw-r--r-- 1 ibm-9 supergroup 0 2026-07-20 09:39 /geeks/cutAndPaste.txt
-rw-r--r-- 1 ibm-9 supergroup 0 2026-07-20 09:31 /geeks/myfile.txt
ibm-9@IBM-9:~/Desktop$
ibm-9@IBM-9:~/Desktop$

```
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -cp /geeks /geeks_copied
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -ls /
Found 2 items
drwxr-xr-x   - ibm-9 supergroup          0 2026-07-20 09:39 /geeks
drwxr-xr-x   - ibm-9 supergroup          0 2026-07-20 09:41 /geeks_copied
ibm-9@IBM-9:~/Desktop$
```

```
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -mv /geeks/myfile.txt /geeks_copie
d
mv: `/geeks_copied/myfile.txt': File exists
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -ls /geeks_copied
Found 3 items
-rw-r--r--   1 ibm-9 supergroup         15 2026-07-20 09:41 /geeks_copied/AI.txt
-rw-r--r--   1 ibm-9 supergroup          0 2026-07-20 09:41 /geeks_copied/cutAndPaste.tx
t
-rw-r--r--   1 ibm-9 supergroup          0 2026-07-20 09:41 /geeks_copied/myfile.txt
ibm-9@IBM-9:~/Desktop$
```

```
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -rmr /geeks_copied
rmr: DEPRECATED: Please use '-rm -r' instead.
Deleted /geeks_copied
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -ls /
Found 1 items
drwxr-xr-x   - ibm-9 supergroup          0 2026-07-20 09:39 /geeks
ibm-9@IBM-9:~/Desktop$
```

```
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -du /geeks
15  15  /geeks/AI.txt
0   0   /geeks/cutAndPaste.txt
0   0   /geeks/myfile.txt
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -du -h /geeks
15  15  /geeks/AI.txt
0   0   /geeks/cutAndPaste.txt
0   0   /geeks/myfile.txt
ibm-9@IBM-9:~/Desktop$
```

```
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -dus /geeks
dus: DEPRECATED: Please use 'du -s' instead.
15  15  /geeks
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -du -s /geeks
15  15  /geeks
ibm-9@IBM-9:~/Desktop$
```

```
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -stat /geeks
2026-07-20 09:39:27
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -stat /geeks/AI.txt
2026-07-20 09:35:03
ibm-9@IBM-9:~/Desktop$
```

```
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -setrep -w 6 /geeks/AI.txt
Replication 6 set: /geeks/AI.txt
Waiting for /geeks/AI.txt .........^Cibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/
hdfs dfs -setrep -R 4 //usr/local/hadoop/bin/hdfs dfs -setrep -R 4 /geeks
Replication 4 set: /geeks/AI.txt
Replication 4 set: /geeks/cutAndPaste.txt
Replication 4 set: /geeks/myfile.txt
ibm-9@IBM-9:~/Desktop$
```

```
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs

Usage: hadoop fs [generic options]
        [-appendToFile [-n] <localsrc> ... <dst>]
        [-cat [-ignoreCrc] <src> ...]
        [-checksum [-v] <src> ...]
        [-chgrp [-R] GROUP PATH...]
        [-chmod [-R] <MODE[,MODE]... | OCTALMODE> PATH...]
        [-chown [-R] [OWNER][:[GROUP]] PATH...]
        [-concat <target path> <src path> <src path> ...]
        [-copyFromLocal [-f] [-p] [-l] [-d] [-t <thread count>] [-q <thread pool queue size>] <localsrc> ... <dst>]
        [-copyToLocal [-f] [-p] [-crc] [-ignoreCrc] [-t <thread count>] [-q <thread pool queue size>] <src> ... <localdst>]
        [-count [-q] [-h] [-v] [-t [<storage type>]] [-u] [-x] [-e] [-s] <path> ...]
        [-cp [-f] [-p | -p[topax]] [-d] [-t <thread count>] [-q <thread pool queue size>] <src> ... <dst>]
        [-createSnapshot <snapshotDir> [<snapshotName>]]
        [-deleteSnapshot <snapshotDir> <snapshotName>]
        [-df [-h] [<path> ...]]
        [-du [-s] [-h] [-v] [-x] <path> ...]
        [-expunge [-immediate] [-fs <path>]]
        [-find <path> ... <expression> ...]
        [-get [-f] [-p] [-crc] [-ignoreCrc] [-t <thread count>] [-q <thread pool queue size>] <src> ... <localdst>]
        [-getfacl [-R] <path>]
        [-getfattr [-R] {-n name | -d} [-e en] <path>]
        [-getmerge [-nl] [-skip-empty-file] <src> <localdst>]
        [-head <file>]
        [-help [cmd ...]]
        [-ls [-C] [-d] [-h] [-q] [-R] [-t] [-S] [-r] [-u] [-e] [<path> ...]]
        [-mkdir [-p] <path> ...]
        [-moveFromLocal [-f] [-p] [-l] [-d] <localsrc> ... <dst>]
        [-moveToLocal <src> <localdst>]
        [-mv <src> ... <dst>]
        [-put [-f] [-p] [-l] [-d] [-t <thread count>] [-q <thread pool queue size>] <localsrc> ... <dst>]
        [-renameSnapshot <snapshotDir> <oldName> <newName>]
        [-rm [-f] [-r|-R] [-skipTrash] [-safely] <src> ...]
        [-rmdir [--ignore-fail-on-non-empty] <dir> ...]
        [-setfacl [-R] [{-b|-k} {-m|-x <acl_spec>} <path>]|[--set <acl_spec> <path>]]
        [-setfattr {-n name [-v value] | -x name} <path>]
        [-setrep [-R] [-w] <rep> <path> ...]
        [-stat [format] <path> ...]
        [-tail [-f] [-s <sleep interval>] <file>]
        [-test -[defswrz] <path>]
```

```
        [-mv <src> ... <dst>]
        [-put [-f] [-p] [-l] [-d] [-t <thread count>] [-q <thread pool queue size>] <localsrc> ... <dst>]
        [-renameSnapshot <snapshotDir> <oldName> <newName>]
        [-rm [-f] [-r|-R] [-skipTrash] [-safely] <src> ...]
        [-rmdir [--ignore-fail-on-non-empty] <dir> ...]
        [-setfacl [-R] [{-b|-k} {-m|-x <acl_spec>} <path>]|[--set <acl_spec> <path>]]
        [-setfattr {-n name [-v value] | -x name} <path>]
        [-setrep [-R] [-w] <rep> <path> ...]
        [-stat [format] <path> ...]
        [-tail [-f] [-s <sleep interval>] <file>]
        [-test -[defswrz] <path>]
        [-text [-ignoreCrc] <src> ...]
        [-touch [-a] [-m] [-t TIMESTAMP (yyyyMMdd:HHmmss) ] [-c] <path> ...]
        [-touchz <path> ...]
        [-truncate [-w] <length> <path> ...]
        [-usage [cmd ...]]

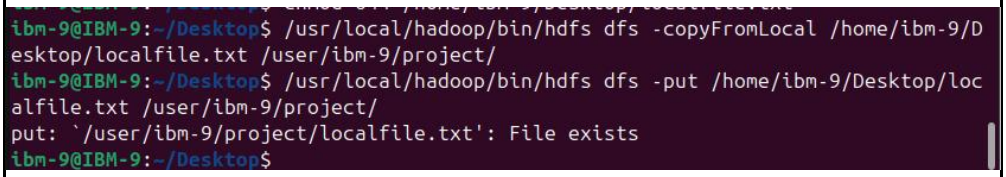
Generic options supported are:
-conf <configuration file>        specify an application configuration file
-D <property=value>               define a value for a given property
-fs <file:///|hdfs://namenode:port> specify default filesystem URL to use, overrides 'fs.defaultFS' property from configurations.
-jt <local|resourcemanager:port>  specify a ResourceManager
-files <file1,...>                specify a comma-separated list of files to be copied to the map reduce cluster
-libjars <jar1,...>               specify a comma-separated list of jar files to be included in the classpath
-archives <archive1,...>          specify a comma-separated list of archives to be unarchived on the compute machines

The general command line syntax is:
command [genericOptions] [commandOptions]

ibm-9@IBM-9:~/Desktop$
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -ls /user/
Found 1 items
drwxr-xr-x   - ibm-9 supergroup          0 2026-07-20 09:52 /user/ibm-9
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -mkdir -p /user/ibm-9/project
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -ls /user/ibm-9/
Found 1 items
drwxr-xr-x   - ibm-9 supergroup          0 2026-07-20 09:52 /user/ibm-9/project
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -chown ibm-9 /user/ibm-9/project/
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -chmod 755 /user/ibm-9/project/
ibm-9@IBM-9:~/Desktop$ /usr/local/hadoop/bin/hdfs dfs -chmod 777 /user/ibm-9/project/
ibm-9@IBM-9:~/Desktop$
```

Step 2: Hadoop User Linux Permissions

```
y
ibm-9@IBM-9:~/Desktop$ touch /home/ibm-9/Desktop/localfile.txt
ibm-9@IBM-9:~/Desktop$ ls -l /home/ibm-9/Desktop/localfile.txt
-rw-rw-r-- 1 ibm-9 ibm-9 0 Jul 20 09:55 /home/ibm-9/Desktop/localfile.txt
ibm-9@IBM-9:~/Desktop$ ls -l /home/ibm-9/Desktop/localfile.txt
-rw-rw-r-- 1 ibm-9 ibm-9 0 Jul 20 09:55 /home/ibm-9/Desktop/localfile.txt
ibm-9@IBM-9:~/Desktop$ chmod 644 /home/ibm-9/Desktop/localfile.txt
ibm-9@IBM-9:~/Desktop$
```

	Step 3: Copy Files from Local to HDFS  Step 4: Verify File Upload 
Conclusion:	Students will be able to set up and configure Hadoop cluster in Pseudo Distributed Mode, which helps to simulate a multi-node installation on a single node and also will be able to work with Hadoop HDFS file system using various commands.
References:	Put the reference of resources used to perform this experiment. https://www.geeksforgeeks.org/linux-unix/hdfs-commands/

Experiment No: 2

Name: Navya Nair **Roll No. : 24**

Batch: B **Performance Date : 7/08/26**

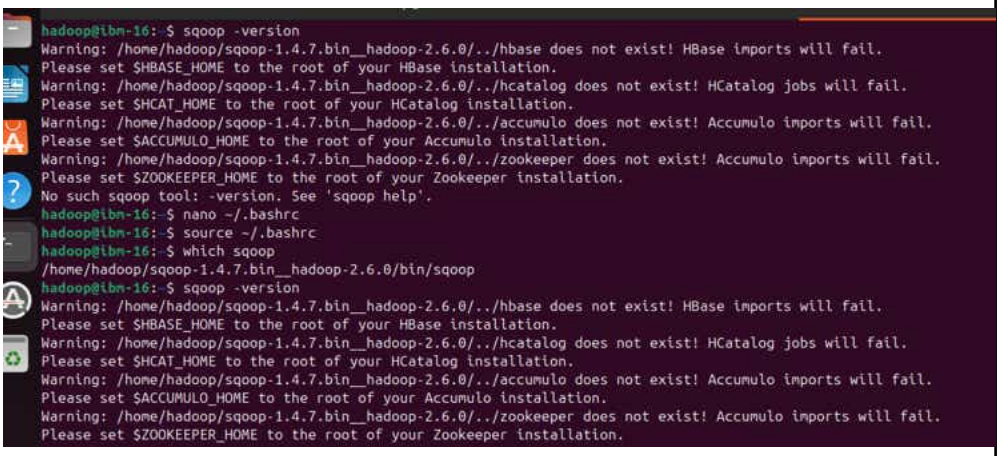
Topic:	Use of Sqoop tool to transfer data between Hadoop and relational database servers. a. Sqoop and MySQL - Installation. b. To execute basic commands of Hadoop eco system component Sqoop.
Prerequisite:	Familiarity with command-line interfaces such as bash Basic knowledge of Relational database management systems MySQL Basic familiarity with the purpose and operation of Hadoop
Mapping With COs:	CSL702.2
Objective:	Ingest data using Sqoop.
Outcomes:	Students will be able to use the Sqoop tool - for transferring data between Hadoop & relational databases.
Instructions:	This experiment is a compulsory experiment. All the students are required to perform this experiment individually.

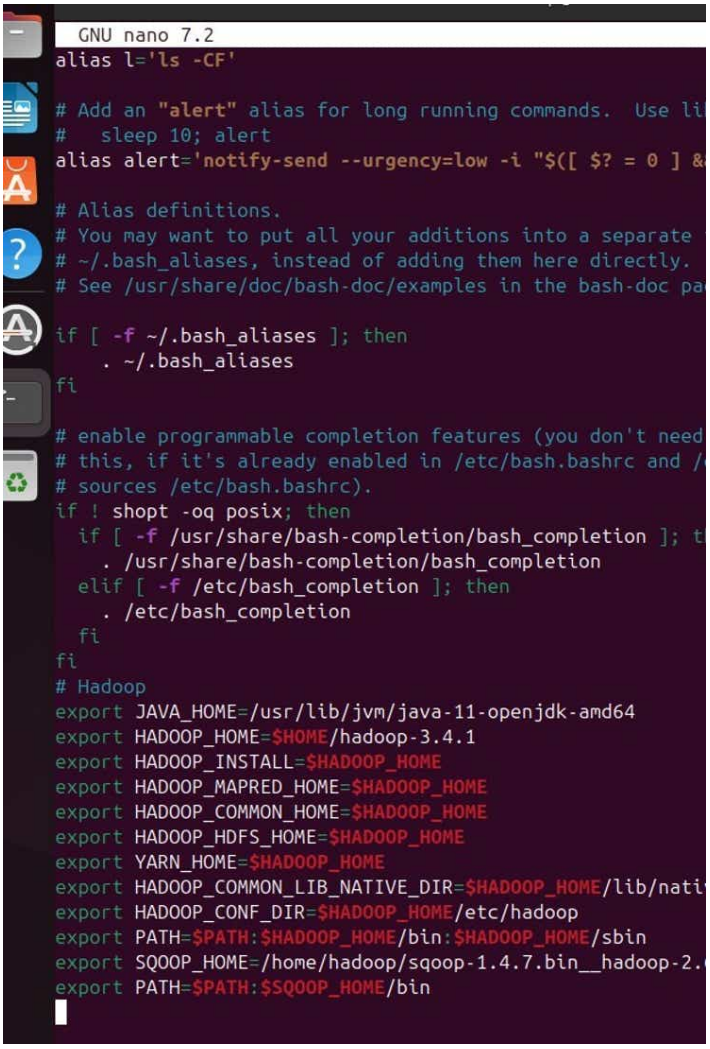
Deliverables:

Submission on Moodle:

Take snapshot of following task:

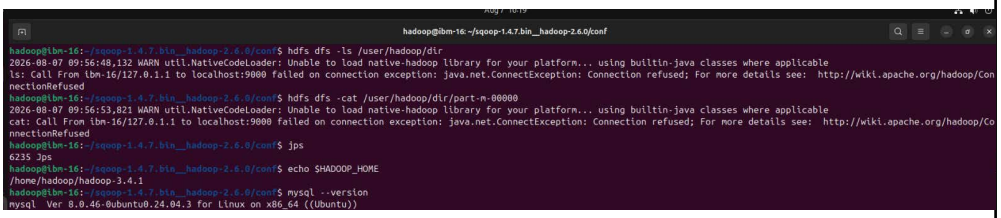
- Sqoop installation

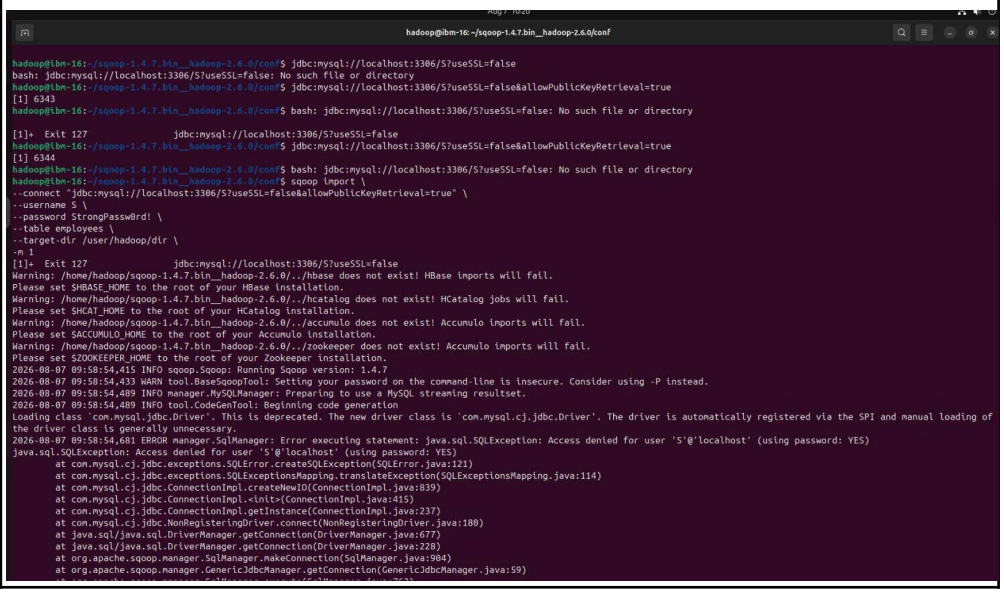


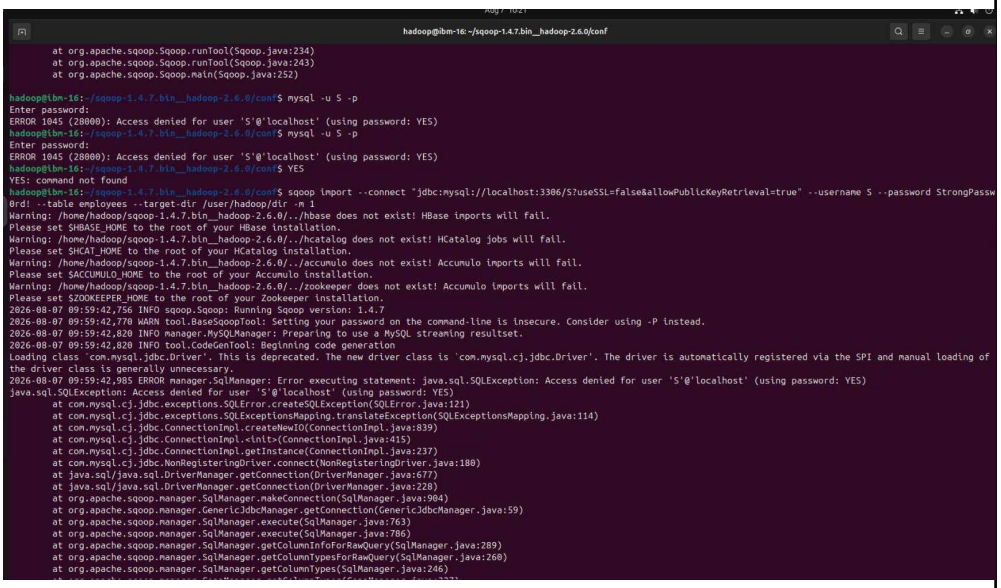


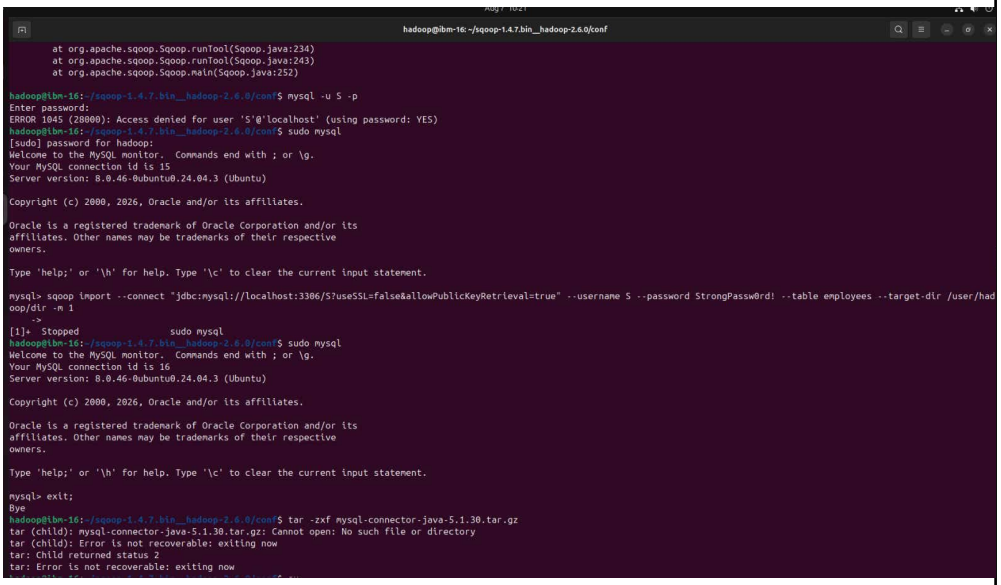
```
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/bin$ sqoop version
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
2026-08-07 09:53:24,841 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
Sqoop 1.4.7
git commit id 2328971411f57f0cb683dfb79d19d4d19d185dd8
Compiled by maugli on Thu Dec 21 15:59:58 STD 2017
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/bin$ cd $SQOOP_HOME/conf
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ ls
oraoop-site-template.xml  sqoop-env-template.cmd  sqoop-env-template.sh  sqoop-site-template.xml  sqoop-site.xml
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ cp sqoop-env-template.sh sqoop-env.sh
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ nano sqoop-env.sh
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ /home/hadoop/mysql-connector-j-9.4.0
bash: /home/hadoop/mysql-connector-j-9.4.0: Is a directory
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ find ~/mysql-connector-j-9.4.0 -name "*.jar"
/home/hadoop/mysql-connector-j-9.4.0/mysql-connector-j-9.4.0.jar
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ cp ~/mysql-connector-j-9.4.0/*.jar $SQOOP_HOME/lib/
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ ls $SQOOP_HOME/lib | grep mysql
mysql-connector-j-9.4.0.jar
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ sqoop version
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
2026-08-07 09:56:13,847 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
Sqoop 1.4.7
git commit id 2328971411f57f0cb683dfb79d19d4d19d185dd8
Compiled by maugli on Thu Dec 21 15:59:58 STD 2017
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ jps
6023 Jps
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ sqoop import \
--connect "jdbc:mysql://localhost:3306/S?useSSL=false" \
--username S \
```

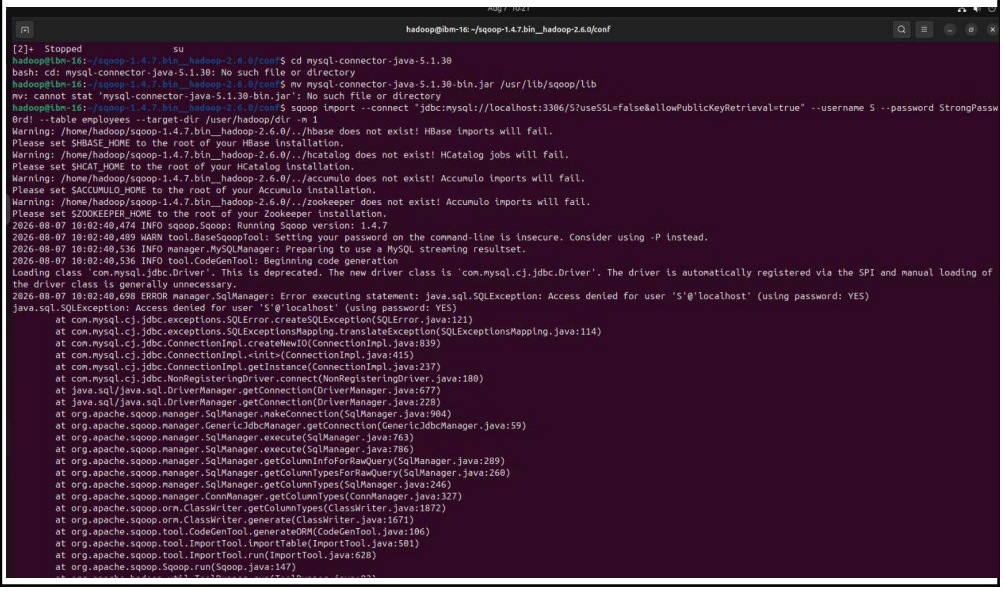
- MySQL installation

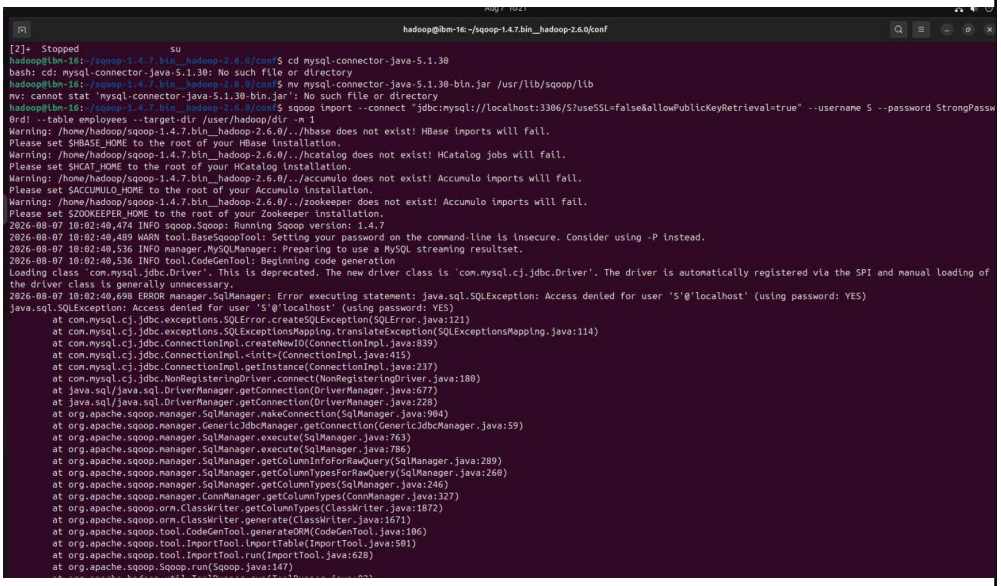




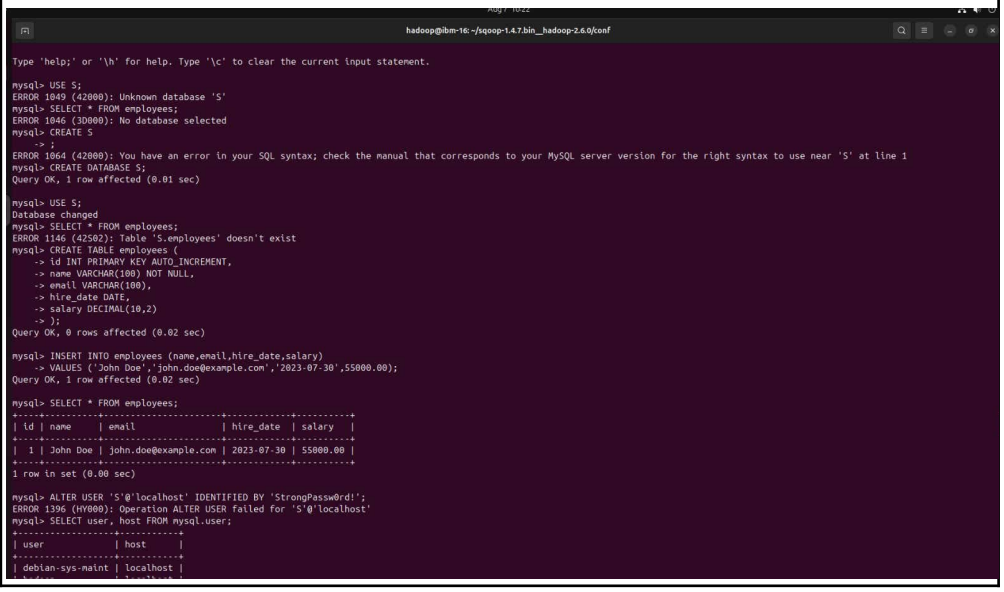

hadoop@ibm-16: ~/sqoop-1.4.7.bin__hadoop-2.6.0/conf
at org.apache.sqoop.Sqoop.runTool(Sqoop.java:234)
at org.apache.sqoop.Sqoop.runTool(Sqoop.java:243)
at org.apache.sqoop.Sqoop.main(Sqoop.java:252)
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ mysql -u S -p
Enter password:
ERROR 1045 (28000): Access denied for user 'S'@'localhost' (using password: YES)
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ mysql -u S -p
Enter password:
ERROR 1045 (28000): Access denied for user 'S'@'localhost' (using password: YES)
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ YES
YES: command not found
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ sqoop import --connect "jdbc:mysql://localhost:3306/S?useSSL=false&allowPublicKeyRetrieval=true" --username S --password StrongPassw
0rd! --table employees --target-dir /user/hadoop/dir -m 1
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
2026-08-07 09:59:42,756 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2026-08-07 09:59:42,770 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2026-08-07 09:59:42,820 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
2026-08-07 09:59:42,820 INFO tool.CodeGenTool: Beginning code generation
Loading class `com.mysql.jdbc.Driver'. This is deprecated. The new driver class is `com.mysql.cj.jdbc.Driver'. The driver is automatically registered via the SPI and manual loading of
the driver class is generally unnecessary.
2026-08-07 09:59:42,985 ERROR manager.SqlManager: Error executing statement: java.sql.SQLException: Access denied for user 'S'@'localhost' (using password: YES)
java.sql.SQLException: Access denied for user 'S'@'localhost' (using password: YES)
at com.mysql.cj.jdbc.exceptions.SQLError.createSQLException(SQLError.java:121)
at com.mysql.cj.jdbc.exceptions.SQLExceptionsMapping.translateException(SQLExceptionsMapping.java:114)
at com.mysql.cj.jdbc.ConnectionImpl.createNewIO(ConnectionImpl.java:839)
at com.mysql.cj.jdbc.ConnectionImpl.<init>(ConnectionImpl.java:415)
at com.mysql.cj.jdbc.ConnectionImpl.getInstance(ConnectionImpl.java:237)
at com.mysql.cj.jdbc.NonRegisteringDriver.connect(NonRegisteringDriver.java:180)
at java.sql/java.sql.DriverManager.getConnection(DriverManager.java:677)
at java.sql/java.sql.DriverManager.getConnection(DriverManager.java:228)
at org.apache.sqoop.manager.SqlManager.makeConnection(SqlManager.java:904)
at org.apache.sqoop.manager.GenericJdbcManager.getConnection(GenericJdbcManager.java:59)
at org.apache.sqoop.manager.SqlManager.execute(SqlManager.java:763)
at org.apache.sqoop.manager.SqlManager.execute(SqlManager.java:786)
at org.apache.sqoop.manager.SqlManager.getColumnInfoForRawQuery(SqlManager.java:289)
at org.apache.sqoop.manager.SqlManager.getColumnTypesForRawQuery(SqlManager.java:260)
at org.apache.sqoop.manager.SqlManager.getColumnTypes(SqlManager.java:246)

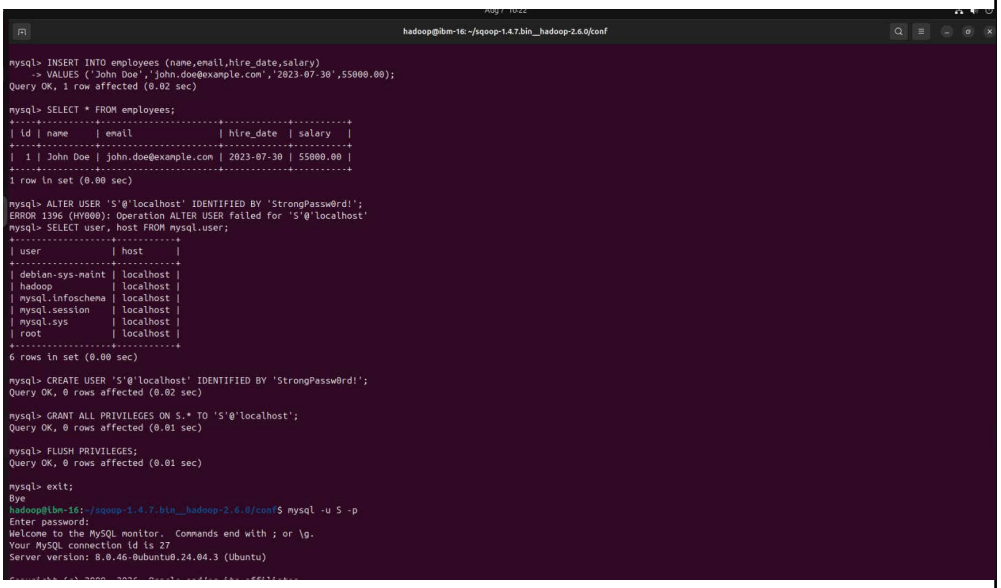

hadoop@ibm-16: ~/sqoop-1.4.7.bin__hadoop-2.6.0/conf
at org.apache.sqoop.Sqoop.runTool(Sqoop.java:234)
at org.apache.sqoop.Sqoop.runTool(Sqoop.java:243)
at org.apache.sqoop.Sqoop.main(Sqoop.java:252)
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ mysql -u S -p
Enter password:
ERROR 1045 (28000): Access denied for user 'S'@'localhost' (using password: YES)
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ sudo mysql
[sudo] password for hadoop:
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 15
Server version: 8.0.46-0ubuntu0.24.04.3 (Ubuntu)
Copyright (c) 2000, 2026, Oracle and/or its affiliates.
Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql> sqoop import --connect "jdbc:mysql://localhost:3306/S?useSSL=false&allowPublicKeyRetrieval=true" --username S --password StrongPassw0rd! --table employees --target-dir /user/had
oop/dir -m 1
-> ^Z
[1]+ Stopped sudo mysql
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ sudo mysql
Welcome to the MySQL monitor. Commands end with ; or \g.
Your MySQL connection id is 16
Server version: 8.0.46-0ubuntu0.24.04.3 (Ubuntu)
Copyright (c) 2000, 2026, Oracle and/or its affiliates.
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affiliates. Other names may be trademarks of their respective
owners.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql> exit;
Bye
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ tar -zxf mysql-connector-java-5.1.30.tar.gz
tar (child): mysql-connector-java-5.1.30.tar.gz: Cannot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now

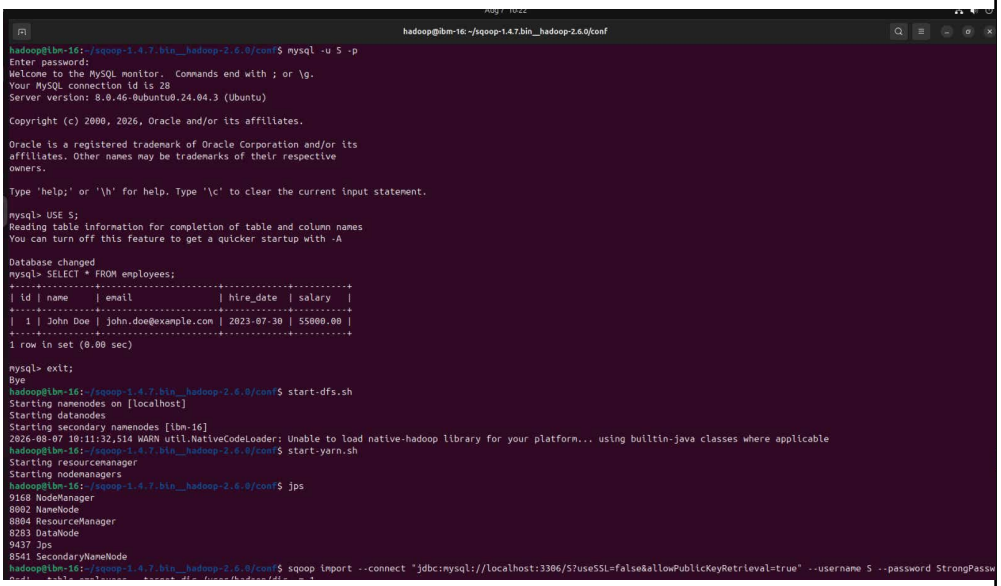

hadoop@ibm-16: ~/sqoop-1.4.7.bin__hadoop-2.6.0/conf
[2]+ Stopped su
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ cd mysql-connector-java-5.1.30
bash: cd: mysql-connector-java-5.1.30: No such file or directory
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ mv mysql-connector-java-5.1.30-bin.jar /usr/lib/sqoop/lib
mv: cannot stat 'mysql-connector-java-5.1.30-bin.jar': No such file or directory
hadoop@ibm-16:~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ sqoop import --connect "jdbc:mysql://localhost:3306/S?useSSL=false&allowPublicKeyRetrieval=true" --username S --password StrongPassw
0rd! --table employees --target-dir /user/hadoop/dir -m 1
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
2026-08-07 10:02:40,474 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2026-08-07 10:02:40,489 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2026-08-07 10:02:40,536 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
2026-08-07 10:02:40,536 INFO tool.CodeGenTool: Beginning code generation
Loading class `com.mysql.jdbc.Driver'. This is deprecated. The new driver class is `com.mysql.cj.jdbc.Driver'. The driver is automatically registered via the SPI and manual loading of
the driver class is generally unnecessary.
2026-08-07 10:02:40,698 ERROR manager.SqlManager: Error executing statement: java.sql.SQLException: Access denied for user 'S'@'localhost' (using password: YES)
java.sql.SQLException: Access denied for user 'S'@'localhost' (using password: YES)
at com.mysql.cj.jdbc.exceptions.SQLError.createSQLException(SQLError.java:121)
at com.mysql.cj.jdbc.exceptions.SQLExceptionsMapping.translateException(SQLExceptionsMapping.java:114)
at com.mysql.cj.jdbc.ConnectionImpl.createNewIO(ConnectionImpl.java:839)
at com.mysql.cj.jdbc.ConnectionImpl.<init>(ConnectionImpl.java:415)
at com.mysql.cj.jdbc.ConnectionImpl.getInstance(ConnectionImpl.java:237)
at com.mysql.cj.jdbc.NonRegisteringDriver.connect(NonRegisteringDriver.java:180)
at java.sql/java.sql.DriverManager.getConnection(DriverManager.java:677)
at java.sql/java.sql.DriverManager.getConnection(DriverManager.java:228)
at org.apache.sqoop.manager.SqlManager.makeConnection(SqlManager.java:904)
at org.apache.sqoop.manager.GenericJdbcManager.getConnection(GenericJdbcManager.java:59)
at org.apache.sqoop.manager.SqlManager.execute(SqlManager.java:763)
at org.apache.sqoop.manager.SqlManager.execute(SqlManager.java:786)
at org.apache.sqoop.manager.SqlManager.getColumnInfoForRawQuery(SqlManager.java:289)
at org.apache.sqoop.manager.SqlManager.getColumnTypesForRawQuery(SqlManager.java:260)
at org.apache.sqoop.manager.SqlManager.getColumnTypes(SqlManager.java:246)
at org.apache.sqoop.manager.ConnManager.getColumnTypes(ConnManager.java:327)
at org.apache.sqoop.orm.ClassWriter.getColumnTypes(ClassWriter.java:1872)
at org.apache.sqoop.orm.ClassWriter.generate(ClassWriter.java:1671)
at org.apache.sqoop.tool.CodeGenTool.generateORM(CodeGenTool.java:106)
at org.apache.sqoop.tool.ImportTool.importTable(ImportTool.java:501)
at org.apache.sqoop.tool.ImportTool.run(ImportTool.java:628)
at org.apache.sqoop.Sqoop.run(Sqoop.java:147)



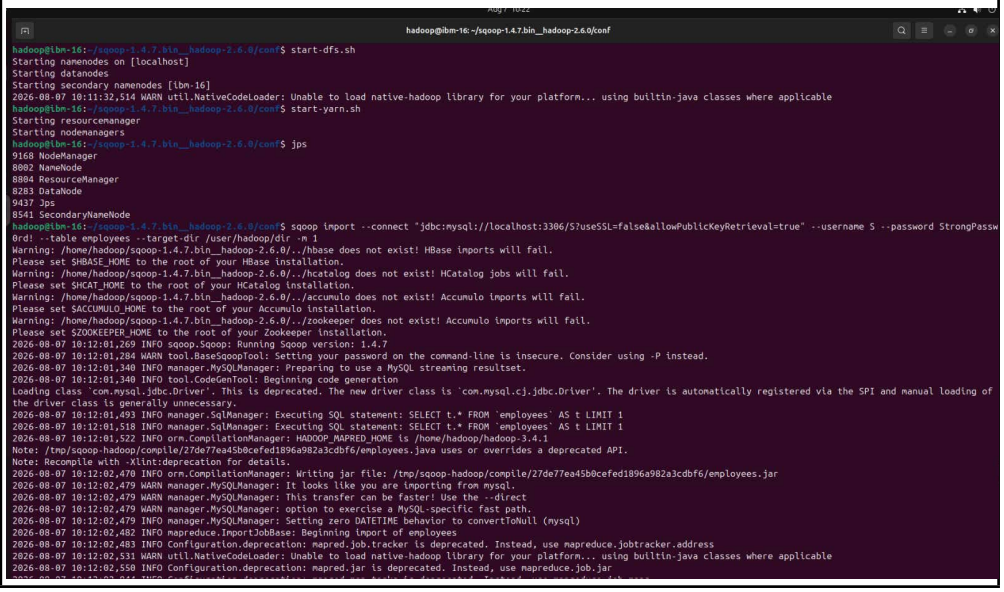
- Database and table creation in MySQL

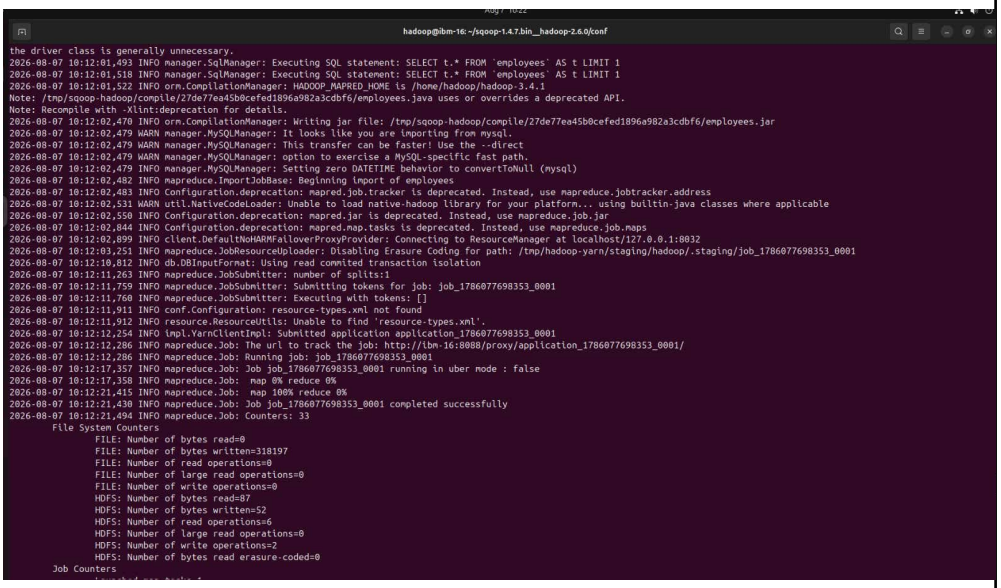




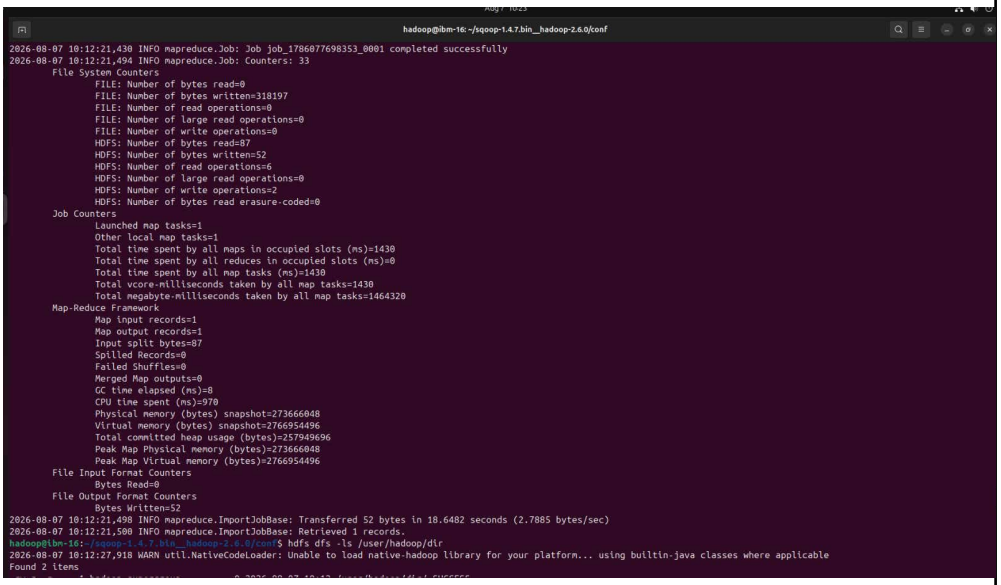


- import (need to work on various types)

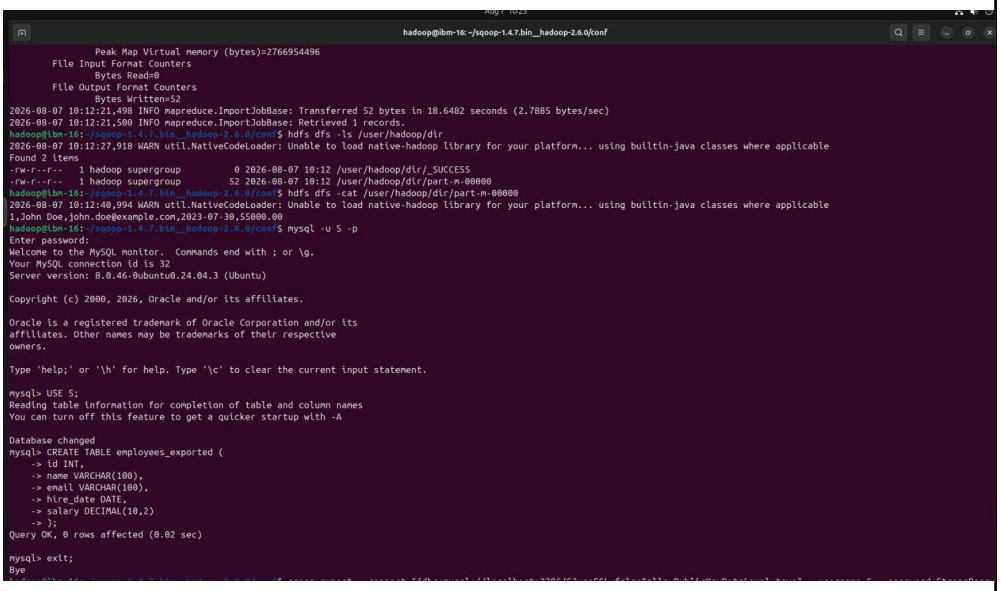




```
hadoop@ibm-16: ~/sqoop-1.4.7.bin__hadoop-2.6.0/conf
the driver class is generally unnecessary.
2026-08-07 10:12:01,493 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM `employees` AS t LIMIT 1
2026-08-07 10:12:01,518 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM `employees` AS t LIMIT 1
2026-08-07 10:12:01,522 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/hadoop/hadoop-3.4.1
Note: /tmp/sqoop-hadoop/compile/27de77ea45b0cefed1896a982a3cdbf6/employees.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2026-08-07 10:12:02,470 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-hadoop/compile/27de77ea45b0cefed1896a982a3cdbf6/employees.jar
2026-08-07 10:12:02,479 WARN manager.MySQLManager: It looks like you are importing from mysql.
2026-08-07 10:12:02,479 WARN manager.MySQLManager: This transfer can be faster! Use the --direct
2026-08-07 10:12:02,479 WARN manager.MySQLManager: option to exercise a MySQL-specific fast path.
2026-08-07 10:12:02,479 INFO manager.MySQLManager: Setting zero DATETIME behavior to convertToNull (mysql)
2026-08-07 10:12:02,482 INFO mapreduce.ImportJobBase: Beginning import of employees
2026-08-07 10:12:02,483 INFO Configuration.deprecation: mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-08-07 10:12:02,531 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-08-07 10:12:02,550 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
2026-08-07 10:12:02,844 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
2026-08-07 10:12:02,899 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at localhost/127.0.0.1:8032
2026-08-07 10:12:03,251 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1786077698353_0001
2026-08-07 10:12:10,812 INFO db.DBInputFormat: Using read commited transaction isolation
2026-08-07 10:12:11,263 INFO mapreduce.JobSubmitter: number of splits:1
2026-08-07 10:12:11,759 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1786077698353_0001
2026-08-07 10:12:11,760 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-08-07 10:12:11,911 INFO conf.Configuration: resource-types.xml not found
2026-08-07 10:12:11,912 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-08-07 10:12:12,254 INFO impl.YarnClientImpl: Submitted applIcation application_1786077698353_0001
2026-08-07 10:12:12,286 INFO mapreduce.Job: The url to track the job: http://ibm-16:8088/proxy/application_1786077698353_0001/
2026-08-07 10:12:12,286 INFO mapreduce.Job: Running job: job_1786077698353_0001
2026-08-07 10:12:17,357 INFO mapreduce.Job: Job job_1786077698353_0001 running in uber mode : false
2026-08-07 10:12:17,358 INFO mapreduce.Job:  map 0% reduce 0%
2026-08-07 10:12:21,415 INFO mapreduce.Job:  map 100% reduce 0%
2026-08-07 10:12:21,430 INFO mapreduce.Job: Job job_1786077698353_0001 completed successfully
2026-08-07 10:12:21,494 INFO mapreduce.Job: Counters: 33
        File System Counters
                FILE: Number of bytes read=0
                FILE: Number of bytes written=318197
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=87
                HDFS: Number of bytes written=52
                HDFS: Number of read operations=6
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
```



```
hadoop@ibm-16: ~/sqoop-1.4.7.bin__hadoop-2.6.0/conf
2026-08-07 10:12:21,430 INFO mapreduce.Job: Job job_1786077698353_0001 completed successfully
2026-08-07 10:12:21,494 INFO mapreduce.Job: Counters: 33
        File System Counters
                FILE: Number of bytes read=0
                FILE: Number of bytes written=318197
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=87
                HDFS: Number of bytes written=52
                HDFS: Number of read operations=6
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=1
                Other local map tasks=1
                Total time spent by all maps in occupied slots (ms)=1430
                Total time spent by all reduces in occupied slots (ms)=0
                Total time spent by all map tasks (ms)=1430
                Total vcore-milliseconds taken by all map tasks=1430
                Total megabyte-milliseconds taken by all map tasks=1464320
        Map-Reduce Framework
                Map input records=1
                Map output records=1
                Input split bytes=87
                Spilled Records=0
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=8
                CPU time spent (ms)=970
                Physical memory (bytes) snapshot=273666048
                Virtual memory (bytes) snapshot=2766954496
                Total committed heap usage (bytes)=257949696
                Peak Map Physical memory (bytes)=273666048
                Peak Map Virtual memory (bytes)=2766954496
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=52
2026-08-07 10:12:21,498 INFO mapreduce.ImportJobBase: Transferred 52 bytes in 18.6482 seconds (2.7885 bytes/sec)
2026-08-07 10:12:21,500 INFO mapreduce.ImportJobBase: Retrieved 1 records.
hadoop@ibm-16: ~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ hdfs dfs -ls /user/hadoop/dir
2026-08-07 10:12:27,918 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
```



```
hadoop@ibm-16: ~/sqoop-1.4.7.bin__hadoop-2.6.0/conf
                Peak Map Virtual memory (bytes)=2766954496
        File Input Format Counters
                Bytes Read=0
        File Output Format Counters
                Bytes Written=52
2026-08-07 10:12:21,498 INFO mapreduce.ImportJobBase: Transferred 52 bytes in 18.6482 seconds (2.7885 bytes/sec)
2026-08-07 10:12:21,500 INFO mapreduce.ImportJobBase: Retrieved 1 records.
hadoop@ibm-16: ~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ hdfs dfs -ls /user/hadoop/dir
2026-08-07 10:12:27,918 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r--   1 hadoop supergroup          0 2026-08-07 10:12 /user/hadoop/dir/_SUCCESS
-rw-r--r--   1 hadoop supergroup         52 2026-08-07 10:12 /user/hadoop/dir/part-m-00000
hadoop@ibm-16: ~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ hdfs dfs -cat /user/hadoop/dir/part-m-00000
2026-08-07 10:12:40,994 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
1,John Doe,john.doe@example.com,2023-07-30,55000.00
hadoop@ibm-16: ~/sqoop-1.4.7.bin__hadoop-2.6.0/conf$ mysql -u S -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 32
Server version: 8.0.46-0ubuntu0.24.04.3 (Ubuntu)

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owners.

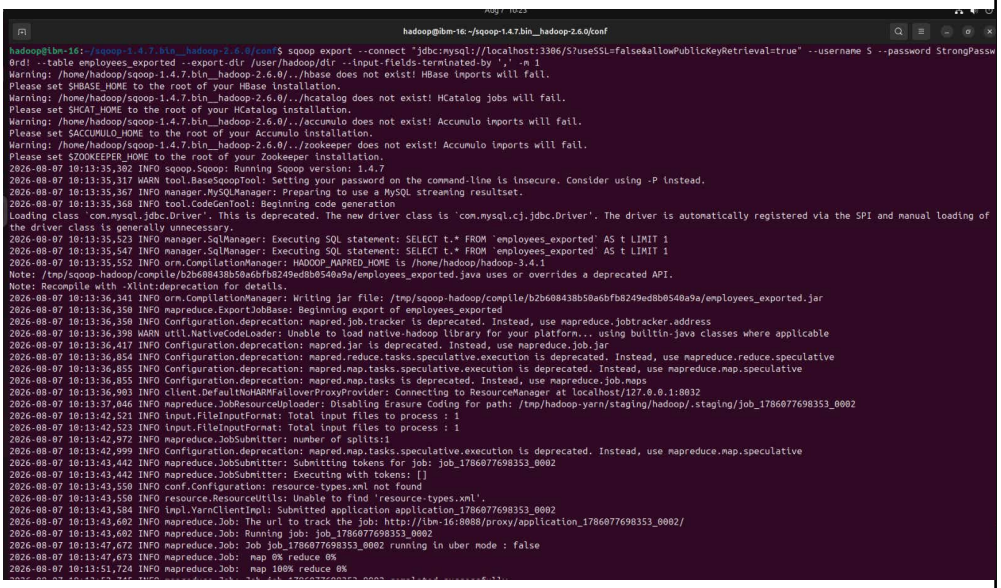
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

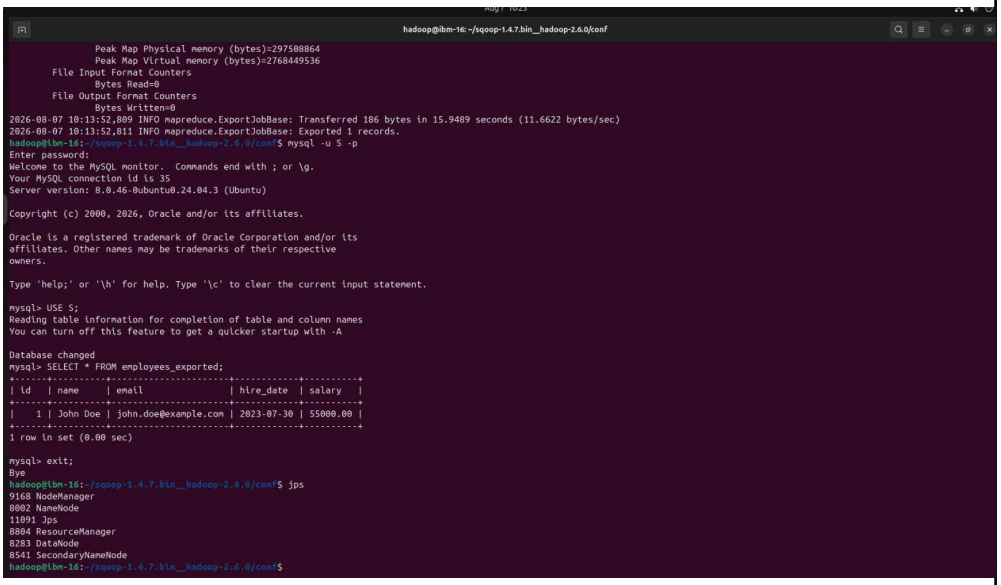
mysql> USE S;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> CREATE TABLE employees_exported (
    -> id INT,
    -> name VARCHAR(100),
    -> email VARCHAR(100),
    -> hire_date DATE,
    -> salary DECIMAL(10,2)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> exit;
Bye
```

- export (need to work on various types)





Exercises for Students:

SPECIFYING A TARGET DIRECTORY:

1. Specify the target directory in HDFS into which we want the output to be saved.

IMPORTING ONLY A SUBSET OF DATA:

```
hadoop@ibm-16:~$ sqoop import \
--connect jdbc:mysql://localhost:3306/college \
--username hadoop \
-P \
--table Faculty \
--target-dir /user/hadoop/faculty_target
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hbase does not exist! HBase import
s will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hcatalog does not exist! HCatalog
jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../accumulo does not exist! Accumulo
imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../zookeeper does not exist! Accumulo
 imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
2026-08-04 10:21:50,793 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
```

```
hadoop@ibm-16:~$ hdfs dfs -ls /user/hadoop/faculty_target
2026-08-04 10:22:25,256 WARN util.NativeCodeLoader: Unable to load native-hadoop library
for your platform... using builtin-java classes where applicable
Found 6 items
-rw-r--r--   1 hadoop supergroup          0 2026-08-04 10:22 /user/hadoop/faculty_target/
_SUCCESS
-rw-r--r--   1 hadoop supergroup         22 2026-08-04 10:22 /user/hadoop/faculty_target/
part-m-00000
-rw-r--r--   1 hadoop supergroup         16 2026-08-04 10:22 /user/hadoop/faculty_target/
part-m-00001
-rw-r--r--   1 hadoop supergroup         19 2026-08-04 10:22 /user/hadoop/faculty_target/
part-m-00002
-rw-r--r--   1 hadoop supergroup         23 2026-08-04 10:22 /user/hadoop/faculty_target/
part-m-00003
-rw-r--r--   1 hadoop supergroup         25 2026-08-04 10:22 /user/hadoop/faculty_target/
part-m-00004
```

```
hadoop@ibm-16:~$ hdfs dfs -cat /user/hadoop/faculty_target/part-m-*
2026-08-04 10:22:32,924 WARN util.NativeCodeLoader: Unable to load native-hadoop library
for your platform... using builtin-java classes where applicable
Computer,1,Amit,60000
IT,2,Neha,55000
EXTC,3,Rahul,50000
Computer,4,Priya,65000
Mechanical,5,Rohit,48000
hadoop@ibm-16:~$
```

2. We will now import only a part of the table Faculty:

PROTECTING YOUR PASSWORD:

```
hadoop@ibm-16:~$ sqoop import \
--connect jdbc:mysql://localhost:3306/college \
--username hadoop \
-P \
--table Faculty \
--where "salary > 55000" \
--target-dir /user/hadoop/faculty_subset
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hbase does not exist! HBase import
s will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hcatalog does not exist! HCatalog
jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../accumulo does not exist! Accumulo
imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../zookeeper does not exist! Accumulo
 imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
2026-08-04 10:24:14,701 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
```

```
2026-08-04 10:24:38,317 INFO mapreduce.ImportJobBase: Retrieved 2 records.
hadoop@ibm-16:~$ hdfs dfs -cat /user/hadoop/faculty_subset/part-m-*
2026-08-04 10:24:41,623 WARN util.NativeCodeLoader: Unable to load native-hadoop library
for your platform... using builtin-java classes where applicable
Computer,1,Amit,60000
Computer,4,Priya,65000
hadoop@ibm-16:~$
```

3. There are two ways of specifying the password. First is to write –P in the command and then specify the password later on. Second way of specifying the password is writing the password in a file and then specifying the file in the command.

Implement both ways.

COMPRESSING IMPORTED DATA:

```
hadoop@ibm-16:~$ sqoop import \
--connect jdbc:mysql://localhost:3306/college \
--username hadoop \
-P \
--table Faculty
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hbase does not exist! HBase import
s will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hcatalog does not exist! HCatalog
jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../accumulo does not exist! Accumulo
imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../zookeeper does not exist! Accumulo
 imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
2026-08-04 10:26:20,052 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
Enter password:
2026-08-04 10:26:33,153 INFO manager.MySQLManager: Preparing to use a MySQL streaming res
```

```
hadoop@ibm-16:~$ echo -n "Hadoop@1234!" > ~/password.txt
hadoop@ibm-16:~$ chmod 400 ~/password.txt
hadoop@ibm-16:~$ sqoop import \
--connect jdbc:mysql://localhost:3306/college \
--username hadoop \
--password-file file:///home/hadoop/password.txt \
--table Faculty \
--target-dir /user/hadoop/faculty_password
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hbase does not exist! HBase import
s will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hcatalog does not exist! HCatalog
jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../accumulo does not exist! Accumulo
imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../zookeeper does not exist! Accumulo
 imports will fail.
```

```
2026-08-04 10:27:31,601 INFO mapreduce.ImportJobBase: Transferred 105 bytes in 17.2375 se
conds (6.0914 bytes/sec)
2026-08-04 10:27:31,603 INFO mapreduce.ImportJobBase: Retrieved 5 records.
hadoop@ibm-16:~$ hdfs dfs -ls /user/hadoop/faculty_password
2026-08-04 10:27:34,135 WARN util.NativeCodeLoader: Unable to load native-hadoop library
for your platform... using builtin-java classes where applicable
Found 6 items
-rw-r--r--   1 hadoop supergroup          0 2026-08-04 10:27 /user/hadoop/faculty_passwor
d/_SUCCESS
-rw-r--r--   1 hadoop supergroup         22 2026-08-04 10:27 /user/hadoop/faculty_passwor
d/part-m-00000
-rw-r--r--   1 hadoop supergroup         16 2026-08-04 10:27 /user/hadoop/faculty_passwor
d/part-m-00001
-rw-r--r--   1 hadoop supergroup         19 2026-08-04 10:27 /user/hadoop/faculty_passwor
d/part-m-00002
-rw-r--r--   1 hadoop supergroup         23 2026-08-04 10:27 /user/hadoop/faculty_passwor
d/part-m-00003
-rw-r--r--   1 hadoop supergroup         25 2026-08-04 10:27 /user/hadoop/faculty_passwor
d/part-m-00004
hadoop@ibm-16:~$
```

4. Use command to compress the imported data.

Exporting DATA FROM HDFS to RELATIONAL DATABASE:

```
d/part-m-00004
hadoop@ibm-16:~$ sqoop import \
--connect jdbc:mysql://localhost:3306/college \
--username hadoop \
-P \
--table Faculty \
--compress \
--target-dir /user/hadoop/faculty_compress
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hbase does not exist! HBase import
s will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../hcatalog does not exist! HCatalog
jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../accumulo does not exist! Accumulo
imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /home/hadoop/sqoop-1.4.7.bin__hadoop-2.6.0/../zookeeper does not exist! Accumulo
 imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
```

```
2026-08-04 10:28:58,936 INFO mapreduce.ImportJobBase: Transferred 201 bytes in 15.043 sec
onds (13.3617 bytes/sec)
2026-08-04 10:28:58,938 INFO mapreduce.ImportJobBase: Retrieved 5 records.
hadoop@ibm-16:~$ hdfs dfs -ls /user/hadoop/faculty_compress
2026-08-04 10:29:02,888 WARN util.NativeCodeLoader: Unable to load native-hadoop library
for your platform... using builtin-java classes where applicable
Found 6 items
-rw-r--r--   1 hadoop supergroup          0 2026-08-04 10:28 /user/hadoop/faculty_compres
s/_SUCCESS
-rw-r--r--   1 hadoop supergroup         40 2026-08-04 10:28 /user/hadoop/faculty_compres
s/part-m-00000.gz
-rw-r--r--   1 hadoop supergroup         36 2026-08-04 10:28 /user/hadoop/faculty_compres
s/part-m-00001.gz
-rw-r--r--   1 hadoop supergroup         37 2026-08-04 10:28 /user/hadoop/faculty_compres
s/part-m-00002.gz
-rw-r--r--   1 hadoop supergroup         43 2026-08-04 10:28 /user/hadoop/faculty_compres
s/part-m-00003.gz
-rw-r--r--   1 hadoop supergroup         45 2026-08-04 10:28 /user/hadoop/faculty_compres
s/part-m-00004.gz
hadoop@ibm-16:~$ hdfs dfs -text /user/hadoop/faculty_compress/part-m-*
```

```
hadoop@ibm-16:~$ hdfs dfs -text /user/hadoop/faculty_compress/part-m-*
2026-08-04 10:29:07,140 WARN util.NativeCodeLoader: Unable to load native-hadoop library
for your platform... using builtin-java classes where applicable
Computer,1,Amit,60000
IT,2,Neha,55000
EXTC,3,Rahul,50000
Computer,4,Priya,65000
Mechanical,5,Rohit,48000
hadoop@ibm-16:~$
```

5. Export the data

```
hadoop@ibm-16:~$ mysql -u hadoop -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 55
Server version: 8.0.46-0ubuntu0.24.04.3 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE college;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql>
mysql> CREATE TABLE Faculty_export (
    ->     department VARCHAR(50),
    ->     id INT,
    ->     name VARCHAR(50),
    ->     salary INT
    -> );
ERROR 1050 (42S01): Table 'Faculty_export' already exists
mysql> EXIT;
Bye
```

```
2026-08-04 10:31:20,788 INFO mapreduce.ExportJobBase: Transferred 667 bytes in
2026-08-04 10:31:20,789 INFO mapreduce.ExportJobBase: Exported 5 records.
hadoop@ibm-16:~$ mysql -u hadoop -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 61
Server version: 8.0.46-0ubuntu0.24.04.3 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE college;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql>
mysql> SELECT * FROM Faculty_export;
+------------+------+-------+--------+
| department | id   | name  | salary |
+------------+------+-------+--------+
| Mechanical |    5 | Rohit |  48000 |
| Computer   |    1 | Amit  |  60000 |
| IT         |    2 | Neha  |  55000 |
| EXTC       |    3 | Rahul |  50000 |
| Computer   |    4 | Priya |  65000 |
| Computer   |    1 | Amit  |  60000 |
| IT         |    2 | Neha  |  55000 |
| Mechanical |    5 | Rohit |  48000 |
| EXTC       |    3 | Rahul |  50000 |
| Computer   |    4 | Priya |  65000 |
+------------+------+-------+--------+
10 rows in set (0.00 sec)

mysql>
```

Conclusion:	Students will be able to use Sqoop tool for transferring data between Hadoop & relational databases.
References:	https://www.tutorialspoint.com/sqoop/sqoop_installation.htm

Experiment No: 3

Name: Navya Nair **Roll No. : 24**

Batch: B **Performance Date : 7-08-2026**

<table>
<tr><td>Topic:</td><td>Implementing programs using MapReduce.a) Implementation of wordcount and total WordCount using MapReduce b) Implementation of Matrix Multiplication using MapReduce</td></tr>
<tr><td>Prerequisite:</td><td>Basic Java programming, Hadoop and MapReduce knowledge is required.</td></tr>
<tr><td>Mapping With COs:</td><td>CSL702.3</td></tr>
<tr><td>Objective:</td><td>To write the program for mapper class, reducer class and driver class to find the number of distinct words present in the input file.</td></tr>
<tr><td>Outcomes:</td><td>Students will be able to write the program for mapper class, reducer class and driver class to find the number of distinct words present in the input file and able to produce part-r file as output.</td></tr>
<tr><td>Instructions:</td><td>This experiment is a compulsory experiments. All the students are required to perform this experiment individually.</td></tr>
<tr><td>Deliverables:</td><td>Submission:

1. Give the details of the input file for this program.
<table>
<tr><td>Input File

1) input.txt</td><td>Link

created locally</td><td>Content detail

Text file containing 4 lines of text used for WordCount and Total WordCount</td><td>File size

110 bytes</td></tr>
<tr><td>2) input2.txt</td><td>created locally</td><td>Text file containing matrix elements of A (2×5) and B (5×3) in <code>matrix,row,column,value</code> format</td><td>245 bytes</td></tr>
</table>
2. Write the code of mapper, reducer and driver.
Word Count
1. Mapper</td></tr>
</table>

```
public static class TokenizerMapper
        extends Mapper<Object, Text, Text, IntWritable> {

     private final static IntWritable one = new IntWritable(1);
     private Text word = new Text();

     public void map(Object key, Text value, Context context)
          throws IOException, InterruptedException {

        StringTokenizer itr = new StringTokenizer(value.toString());

        while (itr.hasMoreTokens()) {
          word.set(itr.nextToken());
          context.write(word, one);
        }
     }
  }
```

2. Reducer

```
public static class IntSumReducer
        extends Reducer<Text, IntWritable, Text, IntWritable> {

    private IntWritable result = new IntWritable();

    public void reduce(Text key, Iterable<IntWritable> values,
                       Context context)
          throws IOException, InterruptedException {

        int sum = 0;

        for (IntWritable val : values) {
          sum += val.get();
        }

        result.set(sum);
        context.write(key, result);
    }
  }
```

3. Driver

```
public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();

    Job job = Job.getInstance(conf, "word count");

    job.setJarByClass(WordCount.class);

    job.setMapperClass(TokenizerMapper.class);
    job.setCombinerClass(IntSumReducer.class);
    job.setReducerClass(IntSumReducer.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
```

```
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
  }
```

TotalCount

1.Mapper

```
public static class TokenizerMapper
        extends Mapper<Object, Text, Text, IntWritable> {

    private final static IntWritable one = new IntWritable(1);
    private Text word = new Text();

    public void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

        StringTokenizer itr = new StringTokenizer(value.toString());

        while (itr.hasMoreTokens()) {
            word.set(itr.nextToken());
            context.write(new Text("total words"), one);
        }
    }
  }
```

2. Reducer

```
public static class IntSumReducer
        extends Reducer<Text, IntWritable, Text, IntWritable> {

    private IntWritable result = new IntWritable();

    public void reduce(Text key, Iterable<IntWritable> values,
                       Context context)
            throws IOException, InterruptedException {

        int sum = 0;

        for (IntWritable val : values) {
            sum += val.get();
        }
```

```
        result.set(sum);
        context.write(key, result);
    }
  }
```

3. Driver

```
 public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();

    Job job = Job.getInstance(conf, "totw");

    job.setJarByClass(totw.class);

    job.setMapperClass(TokenizerMapper.class);
    job.setCombinerClass(IntSumReducer.class);
    job.setReducerClass(IntSumReducer.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
  }
```

Matrix Multiplication

Mapper

```
public static class Map
        extends Mapper<LongWritable, Text, Text, Text> {

    public void map(LongWritable key, Text value, Context context)
            throws IOException, InterruptedException {

        Configuration conf = context.getConfiguration();

        int m = Integer.parseInt(conf.get("m"));
        int p = Integer.parseInt(conf.get("p"));

        String line = value.toString();
        String[] indicesAndValue = line.split(",");

        Text outputKey = new Text();
        Text outputValue = new Text();

        if (indicesAndValue[0].equals("A")) {

            for (int k = 0; k < p; k++) {
                outputKey.set(indicesAndValue[1] + "," + k);
                outputValue.set("A," + indicesAndValue[2]
                        + "," + indicesAndValue[3]);
```

```
                context.write(outputKey, outputValue);
            }

        } else {

            for (int i = 0; i < m; i++) {
                outputKey.set(i + "," + indicesAndValue[2]);
                outputValue.set("B," + indicesAndValue[1]
                        + "," + indicesAndValue[3]);

                context.write(outputKey, outputValue);
            }
        }
    }
}
```

Reducer

```
public static class Reduce
        extends Reducer<Text, Text, Text, Text> {

    public void reduce(Text key, Iterable<Text> values,
                Context context)
            throws IOException, InterruptedException {

        HashMap<Integer, Float> hashA =
                new HashMap<Integer, Float>();

        HashMap<Integer, Float> hashB =
                new HashMap<Integer, Float>();

        for (Text val : values) {

            String[] value = val.toString().split(",");

            if (value[0].equals("A")) {
                hashA.put(
                    Integer.parseInt(value[1]),
                    Float.parseFloat(value[2])
                );
            } else {
                hashB.put(
                    Integer.parseInt(value[1]),
                    Float.parseFloat(value[2])
                );
            }
        }

        int n = Integer.parseInt(
                context.getConfiguration().get("n"));

        float result = 0.0f;
```

```
        for (int j = 0; j < n; j++) {

            float a_ij = hashA.containsKey(j)
                    ? hashA.get(j) : 0.0f;

            float b_jk = hashB.containsKey(j)
                    ? hashB.get(j) : 0.0f;

            result += a_ij * b_jk;
        }

        if (result != 0.0f) {
            context.write(
                null,
                new Text(key.toString() + ","
                        + Float.toString(result))
            );
        }
    }
}
```

Driver

```
public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();

    // A is a 2 x 5 matrix
    // B is a 5 x 3 matrix

    conf.set("m", "2");
    conf.set("n", "5");
    conf.set("p", "3");

    Job job = new Job(
            conf,
            "MatrixMatrixMultiplicationOneStep"
    );

    job.setJarByClass(mm.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(Text.class);

    job.setMapperClass(Map.class);
    job.setReducerClass(Reduce.class);
```

```
    job.setInputFormatClass(TextInputFormat.class);
    job.setOutputFormatClass(TextOutputFormat.class);

    FileInputFormat.addInputPath(
          job, new Path(args[0]));

    FileOutputFormat.setOutputPath(
          job, new Path(args[1]));

    job.waitForCompletion(true);
  }
```

3. Output for this program.(Snapshots)

WordCount

```
hadoop@ibm-16:~$ cd ~
hadoop@ibm-16:~$ nano input.txt
hadoop@ibm-16:~$ cat input.txt
Hadoop is a framework
Hadoop is used for big data
MapReduce is a programming model
Hadoop supports MapReduce
```

```
hadoop@ibm-16:~$ hadoop fs -rm -r -f /input
hadoop@ibm-16:~$ hadoop fs -rm -r -f /output
hadoop@ibm-16:~$ hadoop fs -mkdir -p /input
hadoop@ibm-16:~$ hadoop fs -put input.txt /input
hadoop@ibm-16:~$ hadoop fs -ls /input
Found 1 items
-rw-r--r--   1 hadoop supergroup        110 2026-08-11 09:34 /input/input.txt
```

```
hadoop@ibm-16:~$ cat WordCount.java
import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class WordCount {

    public static class TokenizerMapper
            extends Mapper<Object, Text, Text, IntWritable> {

        private final static IntWritable one = new IntWritable(1);
        private Text word = new Text();

        public void map(Object key, Text value, Context context)
                throws IOException, InterruptedException {

            StringTokenizer itr = new StringTokenizer(value.toString());

            while (itr.hasMoreTokens()) {
                word.set(itr.nextToken());
                context.write(word, one);
            }
        }
    }

    public static class IntSumReducer
            extends Reducer<Text, IntWritable, Text, IntWritable> {

        private IntWritable result = new IntWritable();

        public void reduce(Text key, Iterable<IntWritable> values,
```

```
        public void reduce(Text key, Iterable<IntWritable> values,
                           Context context)
                throws IOException, InterruptedException {

            int sum = 0;

            for (IntWritable val : values) {
                sum += val.get();
            }

            result.set(sum);
            context.write(key, result);
        }
    }

    public static void main(String[] args) throws Exception {

        Configuration conf = new Configuration();

        Job job = Job.getInstance(conf, "word count");

        job.setJarByClass(WordCount.class);

        job.setMapperClass(TokenizerMapper.class);
        job.setCombinerClass(IntSumReducer.class);
        job.setReducerClass(IntSumReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

```
hadoop@ibm-16:~$ javac -classpath "$(hadoop classpath)" -d . WordCount.java
hadoop@ibm-16:~$ find . -name "*.class"
./wordcount/WordCountMapper.class
./wordcount/WordCountReducer.class
./wordcount/WordReducer.class
./wordcount/WordMapper.class
./wordcount/WordCountDriver.class
./WordCount$TokenizerMapper.class
./WordCount.class
./WordCount$IntSumReducer.class
hadoop@ibm-16:~$ jar -cvf WordCount.jar WordCount*.class
added manifest
adding: WordCount$IntSumReducer.class(in = 1755) (out= 749)(deflated 57%)
adding: WordCount$TokenizerMapper.class(in = 1752) (out= 765)(deflated 56%)
adding: WordCount.class(in = 1511) (out= 832)(deflated 44%)
hadoop@ibm-16:~$ ls -l WordCount.jar
-rw-rw-r-- 1 hadoop hadoop 3100 Aug 11 09:36 WordCount.jar
```

```
hadoop@ibm-16:~$ hadoop fs -cat /output/part-r-00000
Hadoop  3
MapReduce       2
a       2
big     1
data    1
for     1
framework       1
is      3
model   1
programming     1
supports        1
used    1
hadoop@ibm-16:~$
```

TotalCount

```
hadoop@ibm-16:~$ nano totw.java
hadoop@ibm-16:~$ cat totw.java
import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class totw {

    public static class TokenizerMapper
            extends Mapper<Object, Text, Text, IntWritable> {

        private final static IntWritable one = new IntWritable(1);
        private Text word = new Text();

        public void map(Object key, Text value, Context context)
                throws IOException, InterruptedException {

            StringTokenizer itr = new StringTokenizer(value.toString());

            while (itr.hasMoreTokens()) {
                word.set(itr.nextToken());
                context.write(new Text("total words"), one);
            }
        }
    }

    public static class IntSumReducer
            extends Reducer<Text, IntWritable, Text, IntWritable> {
```

```
            extends Reducer<Text, IntWritable, Text, IntWritable> {

        private IntWritable result = new IntWritable();

        public void reduce(Text key, Iterable<IntWritable> values,
                           Context context)
                throws IOException, InterruptedException {

            int sum = 0;

            for (IntWritable val : values) {
                sum += val.get();
            }

            result.set(sum);
            context.write(key, result);
        }
    }

    public static void main(String[] args) throws Exception {

        Configuration conf = new Configuration();

        Job job = Job.getInstance(conf, "totw");

        job.setJarByClass(totw.class);

        job.setMapperClass(TokenizerMapper.class);
        job.setCombinerClass(IntSumReducer.class);
        job.setReducerClass(IntSumReducer.class);

        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
```

```
hadoop@ibm-16:~$ javac -classpath "$(hadoop classpath)" -d . totw.java
hadoop@ibm-16:~$ ls -l totw*.class
-rw-rw-r-- 1 hadoop hadoop 1740 Aug 11 09:58 'totw$IntSumReducer.class'
-rw-rw-r-- 1 hadoop hadoop 1764 Aug 11 09:58 'totw$TokenizerMapper.class'
-rw-rw-r-- 1 hadoop hadoop 1478 Aug 11 09:58  totw.class
hadoop@ibm-16:~$ jar -cvf totw.jar totw*.class
added manifest
adding: totw$IntSumReducer.class(in = 1740) (out= 742)(deflated 57%)
adding: totw$TokenizerMapper.class(in = 1764) (out= 777)(deflated 55%)
adding: totw.class(in = 1478) (out= 815)(deflated 44%)
hadoop@ibm-16:~$ ls -l totw.jar
-rw-rw-r-- 1 hadoop hadoop 3058 Aug 11 09:59 totw.jar
hadoop@ibm-16:~$ hadoop fs -rm -r /output
Deleted /output
hadoop@ibm-16:~$ hadoop fs -ls /input
Found 1 items
-rw-r--r--   1 hadoop supergroup        110 2026-08-11 09:34 /input/input.txt
hadoop@ibm-16:~$ hadoop jar totw.jar totw /input /output
```

```
hadoop@ibm-16:~$ hadoop jar totw.jar totw /input /output
2026-08-11 09:59:33,598 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to
anager at localhost/127.0.0.1:8032
2026-08-11 09:59:34,202 WARN mapreduce.JobResourceUploader: Hadoop command-line optio
not performed. Implement the Tool interface and execute your application with ToolRun
edy this.
2026-08-11 09:59:34,231 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding
/tmp/hadoop-yarn/staging/hadoop/.staging/job_1786420272134_0002
2026-08-11 09:59:34,756 INFO input.FileInputFormat: Total input files to process : 1
2026-08-11 09:59:34,855 INFO mapreduce.JobSubmitter: number of splits:1
2026-08-11 09:59:35,125 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1
34_0002
2026-08-11 09:59:35,126 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-08-11 09:59:35,428 INFO conf.Configuration: resource-types.xml not found
2026-08-11 09:59:35,428 INFO resource.ResourceUtils: Unable to find 'resource-types.x
2026-08-11 09:59:35,551 INFO impl.YarnClientImpl: Submitted application application_1
34_0002
2026-08-11 09:59:35,607 INFO mapreduce.Job: The url to track the job: http://ibm-16:8
application_1786420272134_0002/
2026-08-11 09:59:35,608 INFO mapreduce.Job: Running job: job_1786420272134_0002
2026-08-11 09:59:43,799 INFO mapreduce.Job: Job job_1786420272134_0002 running in ube
alse
2026-08-11 09:59:43,801 INFO mapreduce.Job:  map 0% reduce 0%
2026-08-11 09:59:49,908 INFO mapreduce.Job:  map 100% reduce 0%
2026-08-11 09:59:54,956 INFO mapreduce.Job:  map 100% reduce 100%
2026-08-11 09:59:55,981 INFO mapreduce.Job: Job job_1786420272134_0002 completed succ
2026-08-11 09:59:56,140 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=24
                FILE: Number of bytes written=617413
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=212
                HDFS: Number of bytes written=15
```

```
hadoop@ibm-16:~$ hadoop fs -cat /output/part-r-00000
total words     18
hadoop@ibm-16:~$
```

Matrix Multiplication

```
hadoop@ibm-16:~$ nano input.txt
hadoop@ibm-16:~$ nano input2.txt
hadoop@ibm-16:~$ cat input2.txt
A,0,0,1.0
A,0,1,1.0
A,0,2,2.0
A,0,3,3.0
A,0,4,4.0
A,1,0,5.0
A,1,1,6.0
A,1,2,7.0
A,1,3,8.0
A,1,4,9.0
B,0,1,1.0
B,0,2,2.0
B,1,0,3.0
B,1,1,4.0
B,1,2,5.0
B,2,0,6.0
B,2,1,7.0
B,2,2,8.0
B,3,0,9.0
B,3,1,10.0
B,3,2,11.0
B,4,0,12.0
B,4,1,13.0
B,4,2,14.0
hadoop@ibm-16:~$ hadoop fs -mkdir /input
```

```
hadoop@ibm-16:~$ hadoop fs -rm -r /output
Deleted /output
hadoop@ibm-16:~$ hadoop jar mm.jar mm /input /output
2026-08-11 10:15:05,108 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceM
anager at localhost/127.0.0.1:8032
2026-08-11 10:15:05,669 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing
not performed. Implement the Tool interface and execute your application with ToolRunner to rem
edy this.
2026-08-11 10:15:05,699 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path:
/tmp/hadoop-yarn/staging/hadoop/.staging/job_1786420272134_0004
2026-08-11 10:15:06,174 INFO input.FileInputFormat: Total input files to process : 1
2026-08-11 10:15:06,268 INFO mapreduce.JobSubmitter: number of splits:1
2026-08-11 10:15:06,526 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_17864202721
34_0004
2026-08-11 10:15:06,526 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-08-11 10:15:06,847 INFO conf.Configuration: resource-types.xml not found
2026-08-11 10:15:06,847 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-08-11 10:15:06,955 INFO impl.YarnClientImpl: Submitted application application_17864202721
34_0004
2026-08-11 10:15:07,006 INFO mapreduce.Job: The url to track the job: http://ibm-16:8088/proxy/
application_1786420272134_0004/
2026-08-11 10:15:07,007 INFO mapreduce.Job: Running job: job_1786420272134_0004
2026-08-11 10:15:15,181 INFO mapreduce.Job: Job job_1786420272134_0004 running in uber mode : f
alse
2026-08-11 10:15:15,183 INFO mapreduce.Job:  map 0% reduce 0%
2026-08-11 10:15:21,286 INFO mapreduce.Job:  map 100% reduce 0%
2026-08-11 10:15:27,344 INFO mapreduce.Job:  map 100% reduce 100%
```

```
hadoop@ibm-16:~$ cat mm.java
import java.io.IOException;
import java.util.HashMap;

import org.apache.hadoop.fs.Path;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.*;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.input.TextInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
import org.apache.hadoop.mapreduce.lib.output.TextOutputFormat;

public class mm {

    public static class Map
            extends Mapper<LongWritable, Text, Text, Text> {

        public void map(LongWritable key, Text value, Context context)
                throws IOException, InterruptedException {

            Configuration conf = context.getConfiguration();

            int m = Integer.parseInt(conf.get("m"));
            int p = Integer.parseInt(conf.get("p"));

            String line = value.toString();
            String[] indicesAndValue = line.split(",");

            Text outputKey = new Text();
            Text outputValue = new Text();

            if (indicesAndValue[0].equals("A")) {

                for (int k = 0; k < p; k++) {
                    outputKey.set(indicesAndValue[1] + "," + k);
                    outputValue.set("A," + indicesAndValue[2]
                            + "," + indicesAndValue[3]);
```

```
        if (indicesAndValue[0].equals("A")) {

            for (int k = 0; k < p; k++) {
                outputKey.set(indicesAndValue[1] + "," + k);
                outputValue.set("A," + indicesAndValue[2]
                        + "," + indicesAndValue[3]);

                context.write(outputKey, outputValue);
            }

        } else {

            for (int i = 0; i < m; i++) {
                outputKey.set(i + "," + indicesAndValue[2]);
                outputValue.set("B," + indicesAndValue[1]
                        + "," + indicesAndValue[3]);

                context.write(outputKey, outputValue);
            }
        }
    }
}

public static class Reduce
        extends Reducer<Text, Text, Text, Text> {

    public void reduce(Text key, Iterable<Text> values,
                       Context context)
            throws IOException, InterruptedException {

        HashMap<Integer, Float> hashA =
                new HashMap<Integer, Float>();

        HashMap<Integer, Float> hashB =
                new HashMap<Integer, Float>();

        for (Text val : values) {

            String[] value = val.toString().split(",");

            if (value[0].equals("A")) {
```

```
HashMap<Integer, Float> hashA =
        new HashMap<Integer, Float>();

HashMap<Integer, Float> hashB =
        new HashMap<Integer, Float>();

for (Text val : values) {

    String[] value = val.toString().split(",");

    if (value[0].equals("A")) {
        hashA.put(
            Integer.parseInt(value[1]),
            Float.parseFloat(value[2])
        );
    } else {
        hashB.put(
            Integer.parseInt(value[1]),
            Float.parseFloat(value[2])
        );
    }
}

int n = Integer.parseInt(
        context.getConfiguration().get("n"));

float result = 0.0f;

for (int j = 0; j < n; j++) {

    float a_ij = hashA.containsKey(j)
            ? hashA.get(j) : 0.0f;

    float b_jk = hashB.containsKey(j)
            ? hashB.get(j) : 0.0f;

    result += a_ij * b_jk;
}

if (result != 0.0f) {
    context.write(
```

```
            context.write(
                null,
                new Text(key.toString() + ","
                        + Float.toString(result))
            );
        }
    }
}

public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();

    // A is a 2 x 5 matrix
    // B is a 5 x 3 matrix

    conf.set("m", "2");
    conf.set("n", "5");
    conf.set("p", "3");

    Job job = new Job(
            conf,
            "MatrixMatrixMultiplicationOneStep"
    );

    job.setJarByClass(mm.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(Text.class);

    job.setMapperClass(Map.class);
    job.setReducerClass(Reduce.class);

    job.setInputFormatClass(TextInputFormat.class);
    job.setOutputFormatClass(TextOutputFormat.class);

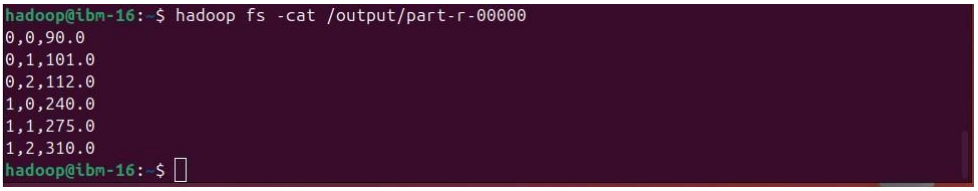
    FileInputFormat.addInputPath(
            job, new Path(args[0]));

    FileOutputFormat.setOutputPath(
            job, new Path(args[1]));
```

```
    FileInputFormat.addInputPath(
            job, new Path(args[0]));

    FileOutputFormat.setOutputPath(
            job, new Path(args[1]));

    job.waitForCompletion(true);
}
```

	 All above 3 steps should be uploaded in a single file in pdf file format on Moodle.
Conclusion:	Thus we are able to execute the Mapreduce program to count distinct words in the input file using hadoop pseudo distributed mode.
References:	1. https://www.geeksforgeeks.org/data-engineering/how-to-execute-wordcount-program-in-mapreduce-using-cloudera-distribution-hadoop-cdh/ 2. https://docs.google.com/document/d/1wXPUiDD3Sg16jVYHcsaly78m3bcsfbVoyu5HxTiTOrk/edit?usp=sharing

Experiment No: 4

Name: Navya Nair **Roll No. : 24**

Class Batch: BE B- B **Performance Date :12/8/26**

Topic:	To install and configure MongoDB to execute NoSQL commands.
Prerequisite:	Basic knowledge of database, SQL, NoSQL
Mapping With COs:	CSL702.5
Objective:	Able to collect, manage, store, query and analyze various forms of Big Data.
Outcomes:	Students will be able to install and manage big data using NoSQL database (Mongodb).
Deliverables:	**Installation**  **MongoDB Operations – Hospital Management System** 1. **Create Database** 2. **Create Collection** 3. **Insert One Patient** 4. **Insert Multiple Patients** 5. **Display All Patients** 6. **Find Patient by Patient ID** 7. **Find Patients by Disease** 8. **Find Patients by Fees** 9. **AND Condition** 10. **OR Condition** 11. **Projection** 12. **Sort Patients by Fees** 13. **Update One Patient** 14. **Update Multiple Patients** 15. **Increase Patient Fees** 16. **Delete One Patient**

17. **Delete Patients by Fees**
18. **Display Remaining Patients**
19. **Count Total Patients**
20. **Exit MongoDB Shell**

Output:

```
test> use hospitalDB
switched to db hospitalDB
hospitalDB> db.createCollection("patients")
{ ok: 1 }
hospitalDB> db.patients.insertOne({
|   patientId: 101,
|   name: "Rahul",
|   age: 25,
|   gender: "Male",
|   disease: "Fever",
|   doctor: "Dr. Sharma",
|   fees: 500,
|   admitted: true
| })
{
  acknowledged: true,
  insertedId: ObjectId('6a7bf51b0a1b6fcf20ef77f5')
}
hospitalDB> db.patients.insertMany([
|   {
|     patientId: 102,
|     name: "Priya",
|     age: 32,
|     gender: "Female",
|     disease: "Diabetes",
|     doctor: "Dr. Mehta",
|     fees: 1200,
|     admitted: false
|   },
|   {
|     patientId: 103,
|     name: "Amit",
|     age: 45,
|     gender: "Male",
|     disease: "Blood Pressure",
|     doctor: "Dr. Sharma",
```

```
|     doctor: "Dr. Sharma",
|     fees: 1500,
|     admitted: true
|   },
|   {
|     patientId: 104,
|     name: "Sneha",
|     age: 28,
|     gender: "Female",
|     disease: "Migraine",
|     doctor: "Dr. Patel",
|     fees: 800,
|     admitted: false
|   },
|   {
|     patientId: 105,
|     name: "Rohan",
|     age: 60,
|     gender: "Male",
|     disease: "Heart Problem",
|     doctor: "Dr. Mehta",
|     fees: 2500,
|     admitted: true
|   },
|   {
|     patientId: 106,
|     name: "Neha",
|     age: 35,
|     gender: "Female",
|     disease: "Fever",
|     doctor: "Dr. Patel",
|     fees: 600,
|     admitted: false
|   }
| ])
{
  acknowledged: true,
```

```
|     admitted: false
|   }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a7bf5200a1b6fcf20ef77f6'),
    '1': ObjectId('6a7bf5200a1b6fcf20ef77f7'),
    '2': ObjectId('6a7bf5200a1b6fcf20ef77f8'),
    '3': ObjectId('6a7bf5200a1b6fcf20ef77f9'),
    '4': ObjectId('6a7bf5200a1b6fcf20ef77fa')
  }
}
hospitalDB> db.patients.find().pretty()
[
  {
    _id: ObjectId('6a7bf51b0a1b6fcf20ef77f5'),
    patientId: 101,
    name: 'Rahul',
    age: 25,
    gender: 'Male',
    disease: 'Fever',
    doctor: 'Dr. Sharma',
    fees: 500,
    admitted: true
  },
  {
    _id: ObjectId('6a7bf5200a1b6fcf20ef77f6'),
    patientId: 102,
    name: 'Priya',
    age: 32,
    gender: 'Female',
    disease: 'Diabetes',
    doctor: 'Dr. Mehta',
    fees: 1200,
    admitted: false
  },
```

```
},
{
  _id: ObjectId('6a7bf5200a1b6fcf20ef77f7'),
  patientId: 103,
  name: 'Amit',
  age: 45,
  gender: 'Male',
  disease: 'Blood Pressure',
  doctor: 'Dr. Sharma',
  fees: 1500,
  admitted: true
},
{
  _id: ObjectId('6a7bf5200a1b6fcf20ef77f8'),
  patientId: 104,
  name: 'Sneha',
  age: 28,
  gender: 'Female',
  disease: 'Migraine',
  doctor: 'Dr. Patel',
  fees: 800,
  admitted: false
},
{
  _id: ObjectId('6a7bf5200a1b6fcf20ef77f9'),
  patientId: 105,
  name: 'Rohan',
  age: 60,
  gender: 'Male',
  disease: 'Heart Problem',
  doctor: 'Dr. Mehta',
  fees: 2500,
  admitted: true
},
{
  _id: ObjectId('6a7bf5200a1b6fcf20ef77fa'),
  patientId: 106,
```

```
    gender: 'Male',
    disease: 'Heart Problem',
    doctor: 'Dr. Mehta',
    fees: 2500,
    admitted: true
  },
  {
    _id: ObjectId('6a7bf5200a1b6fcf20ef77fa'),
    patientId: 106,
    name: 'Neha',
    age: 35,
    gender: 'Female',
    disease: 'Fever',
    doctor: 'Dr. Patel',
    fees: 600,
    admitted: false
  }
]
hospitalDB> db.patients.find({
|   patientId: 103
| })
[
  {
    _id: ObjectId('6a7bf5200a1b6fcf20ef77f7'),
    patientId: 103,
    name: 'Amit',
    age: 45,
    gender: 'Male',
    disease: 'Blood Pressure',
    doctor: 'Dr. Sharma',
    fees: 1500,
    admitted: true
  }
]
hospitalDB> db.patients.find({
|   disease: "Fever"
| })
```

```
  { name: 'Sneha', disease: 'Migraine' },
  { name: 'Rohan', disease: 'Heart Problem' },
  { name: 'Neha', disease: 'Fever' }
]
hospitalDB> db.patients.find().sort({
|   fees: -1
| })
[
  {
    _id: ObjectId('6a7bf5200a1b6fcf20ef77f9'),
    patientId: 105,
    name: 'Rohan',
    age: 60,
    gender: 'Male',
    disease: 'Heart Problem',
    doctor: 'Dr. Mehta',
    fees: 2500,
    admitted: true
  },
  {
    _id: ObjectId('6a7bf5200a1b6fcf20ef77f7'),
    patientId: 103,
    name: 'Amit',
    age: 45,
    gender: 'Male',
    disease: 'Blood Pressure',
    doctor: 'Dr. Sharma',
    fees: 1500,
    admitted: true
  },
  {
    _id: ObjectId('6a7bf5200a1b6fcf20ef77f6'),
    patientId: 102,
    name: 'Priya',
    age: 32,
    gender: 'Female',
    disease: 'Diabetes',
```

```
hospitalDB> db.patients.updateOne(
|   { patientId: 101 },
|   { $set: { fees: 700 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
hospitalDB> db.patients.updateMany(
|   { disease: "Fever" },
|   { $set: { admitted: true } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 1,
  upsertedCount: 0
}
hospitalDB> db.patients.updateOne(
|   { patientId: 104 },
|   { $inc: { fees: 100 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
hospitalDB> db.patients.deleteOne({
```

```
hospitalDB> db.patients.deleteOne({
|   patientId: 105
| })
{ acknowledged: true, deletedCount: 1 }
hospitalDB> db.patients.deleteMany({
|   fees: { $lt: 700 }
| })
{ acknowledged: true, deletedCount: 1 }
hospitalDB> db.patients.find().pretty()
[
  {
    _id: ObjectId('6a7bf51b0a1b6fcf20ef77f5'),
    patientId: 101,
    name: 'Rahul',
    age: 25,
    gender: 'Male',
    disease: 'Fever',
    doctor: 'Dr. Sharma',
    fees: 700,
    admitted: true
  },
  {
    _id: ObjectId('6a7bf5200a1b6fcf20ef77f6'),
    patientId: 102,
    name: 'Priya',
    age: 32,
    gender: 'Female',
    disease: 'Diabetes',
    doctor: 'Dr. Mehta',
    fees: 1200,
    admitted: false
  },
  {
    _id: ObjectId('6a7bf5200a1b6fcf20ef77f7'),
    patientId: 103,
    name: 'Amit',
    age: 45,
```

<table>
<tr><td></td><td>

```
    _id: ObjectId('6a7bf5200a1b6fcf20ef77f6'),
    patientId: 102,
    name: 'Priya',
    age: 32,
    gender: 'Female',
    disease: 'Diabetes',
    doctor: 'Dr. Mehta',
    fees: 1200,
    admitted: false
  },
  {
    _id: ObjectId('6a7bf5200a1b6fcf20ef77f7'),
    patientId: 103,
    name: 'Amit',
    age: 45,
    gender: 'Male',
    disease: 'Blood Pressure',
    doctor: 'Dr. Sharma',
    fees: 1500,
    admitted: true
  },
  {
    _id: ObjectId('6a7bf5200a1b6fcf20ef77f8'),
    patientId: 104,
    name: 'Sneha',
    age: 28,
    gender: 'Female',
    disease: 'Migraine',
    doctor: 'Dr. Patel',
    fees: 900,
    admitted: false
  }
]
hospitalDB> db.patients.countDocuments()
4
hospitalDB> exit
complab3-16@complab3-16:~$
```

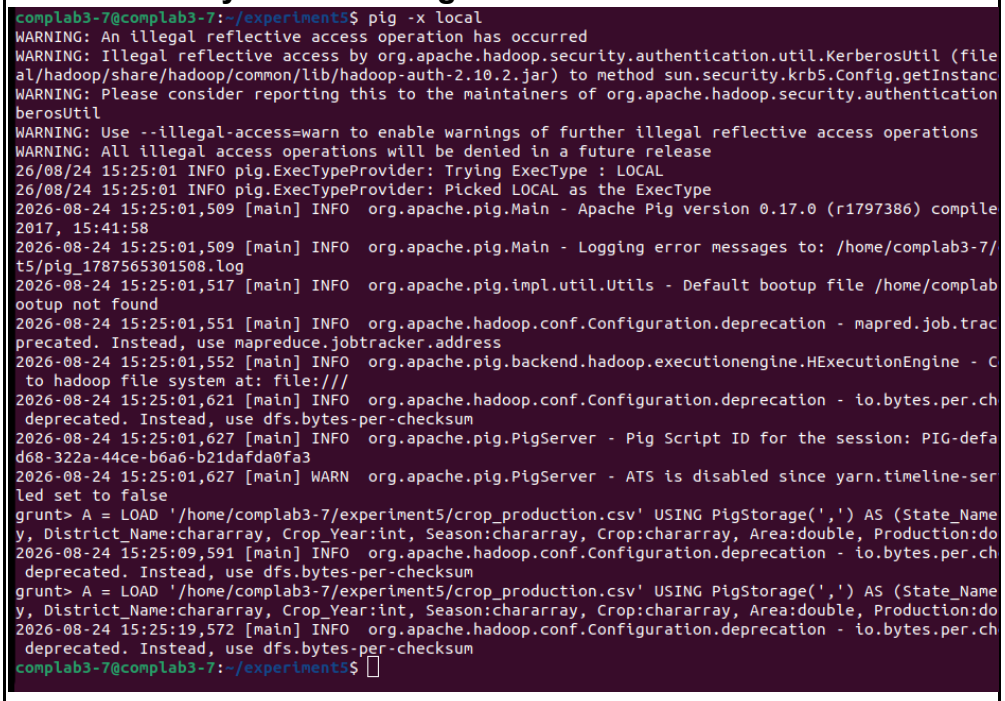
</td></tr>
<tr><td>Conclusion:</td><td>Thus students will be able to successfully install mongodb and apply CRUD operations to manage the big data.</td></tr>
<tr><td>References:</td><td>1. https://medium.com/@sushilanildubey/tutorial-getting-started with-mongodb-5e9e4ab4feef
2. https://www.mongodb.com/docs/manual/crud/</td></tr>
</table>

Lab Manual - BDA

Experiment No: 5

Name: Navya Nair **Roll No. : 24**

Batch: B **Performance Date :08/09/26**

Topic:	To install, configure and execute Apache PIG Latin commands on sample datasets.
Prerequisite:	Basic knowledge of database, PIG Latin language
Mapping With COs:	**CSL702.4**
Objective:	Able to collect, manage, store, query and analyze various forms of Big Data.
Outcomes:	Students will be able to learn, practice and examine tremendous datasets stored in the *HDFS/Local* using PIG scripting tool. Also experiment with Apache Pig and Pig Latin to simplify executing Hadoop MapReduce programs.
Instructions:	This experiment is a compulsory experiment. All the students are required to perform this experiment individually.
Deliverables:	**Part 1:** 1. Download the dataset containing the Agriculture related data about crops in various regions and their area and produce. crop_production.csv (Dataset available on Moodle) **Query 1: Load the agriculture dataset**  Solve following queries to analyze big data : (Capture all queries along with its output).

```
complab3-7@complab3-7:~/experiment5$ pig -x local
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.hadoop.security.authentication.util.KerberosUtil (file
al/hadoop/share/hadoop/common/lib/hadoop-auth-2.10.2.jar) to method sun.security.krb5.Config.getInstanc
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.security.authentication
berosUtil
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
26/08/24 15:25:01 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
26/08/24 15:25:01 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-08-24 15:25:01,509 [main] INFO  org.apache.pig.Main - Apache Pig version 0.17.0 (r1797386) compile
2017, 15:41:58
2026-08-24 15:25:01,509 [main] INFO  org.apache.pig.Main - Logging error messages to: /home/complab3-7/
t5/pig_1787565301508.log
2026-08-24 15:25:01,517 [main] INFO  org.apache.pig.impl.util.Utils - Default bootup file /home/complab
ootup not found
2026-08-24 15:25:01,551 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - mapred.job.trac
precated. Instead, use mapreduce.jobtracker.address
2026-08-24 15:25:01,552 [main] INFO  org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - C
 to hadoop file system at: file:///
2026-08-24 15:25:01,621 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.ch
 deprecated. Instead, use dfs.bytes-per-checksum
2026-08-24 15:25:01,627 [main] INFO  org.apache.pig.PigServer - Pig Script ID for the session: PIG-defa
d68-322a-44ce-b6a6-b21dafda0fa3
2026-08-24 15:25:01,627 [main] WARN  org.apache.pig.PigServer - ATS is disabled since yarn.timeline-ser
led set to false
grunt> A = LOAD '/home/complab3-7/experiment5/crop_production.csv' USING PigStorage(',') AS (State_Name
y, District_Name:chararray, Crop_Year:int, Season:chararray, Crop:chararray, Area:double, Production:do
2026-08-24 15:25:09,591 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.ch
 deprecated. Instead, use dfs.bytes-per-checksum
grunt> A = LOAD '/home/complab3-7/experiment5/crop_production.csv' USING PigStorage(',') AS (State_Name
y, District_Name:chararray, Crop_Year:int, Season:chararray, Crop:chararray, Area:double, Production:do
2026-08-24 15:25:19,572 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - io.bytes.per.ch
 deprecated. Instead, use dfs.bytes-per-checksum
complab3-7@complab3-7:~/experiment5$
```

Query 2: Dump and describe the data set agriculture

```
Details at logfile: /home/complab3-7/experiment5/pig_1787565491230.log
grunt>
grunt> DESCRIBE A;
2026-08-24 15:28:38,456 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 1003:
rator for alias A
Details at logfile: /home/complab3-7/experiment5/pig_1787565491230.log
grunt> DUMP A;
2026-08-24 15:28:43,144 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 1003:
rator for alias A
Details at logfile: /home/complab3-7/experiment5/pig_1787565491230.log
grunt>
```

Query 3: Grouping All Records State wise

```
grunt> B = GROUP A BY State_Name;
2026-08-24 15:29:17,388 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 1200: Pig scrip
parse: <line 1, column 10> Undefined alias: A
Details at logfile: /home/complab3-7/experiment5/pig_1787565491230.log
grunt> DUMP B;
2026-08-24 15:29:21,815 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 1003: Unable to
rator for alias B
Details at logfile: /home/complab3-7/experiment5/pig_1787565491230.log
grunt>
```

Query 4: Highest produced crops and details from each State

```
grunt> C = GROUP A BY (State_Name, Crop);
2026-08-24 15:30:07,802 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 1200: Pig script fai
parse: <line 1, column 10> Undefined alias: A
Details at logfile: /home/complab3-7/experiment5/pig_1787565491230.log
grunt> D = FOREACH C GENERATE group.State_Name AS State_Name, group.Crop AS Crop, SUM(A.Productior
tal_Production;
2026-08-24 15:30:11,658 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 1200: Pig script fai
parse: <line 1, column 12> Undefined alias: C
Details at logfile: /home/complab3-7/experiment5/pig_1787565491230.log
grunt> E = GROUP D BY State_Name;
2026-08-24 15:30:15,175 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 1200: Pig script fai
parse: <line 1, column 10> Undefined alias: D
Details at logfile: /home/complab3-7/experiment5/pig_1787565491230.log
grunt> F = FOREACH E GENERATE group AS State_Name, MAX(D.Total_Production) AS Max_Production;
2026-08-24 15:30:18,952 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 1200: Pig script fai
parse: <line 1, column 12> Undefined alias: E
Details at logfile: /home/complab3-7/experiment5/pig_1787565491230.log
grunt> G = JOIN D BY (State_Name, Total_Production), F BY (State_Name, Max_Production);
2026-08-24 15:30:24,204 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 1200: Pig script fai
parse: <line 1, column 9> Undefined alias: D
Details at logfile: /home/complab3-7/experiment5/pig_1787565491230.log
grunt> H = FOREACH G GENERATE D::State_Name, D::Crop, D::Total_Production;
2026-08-24 15:30:28,520 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 1200: Pig script fai
parse: <line 1, column 12> Undefined alias: G
Details at logfile: /home/complab3-7/experiment5/pig_1787565491230.log
grunt> DUMP H;
2026-08-24 15:30:35,687 [main] ERROR org.apache.pig.tools.grunt.Grunt - ERROR 1003: Unable to finc
rator for alias H
Details at logfile: /home/complab3-7/experiment5/pig_1787565491230.log
grunt>
```

Query 6: Transfer data_top.txt from Local file syste into HDFS

```
complab3-7@complab3-7:~/experiment5$ hdfs dfs -cat /experiment5/data_top.txt/part-r-0
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.hadoop.security.authentication.util.
usr/local/hadoop/share/hadoop/common/lib/hadoop-auth-2.10.2.jar) to method sun.securi
stance()
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.secur
til.KerberosUtil
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective a
WARNING: All illegal access operations will be denied in a future release
Andaman and Nicobar Islands,Coconut ,5.4983E8
complab3-7@complab3-7:~/experiment5$
```

Part 2

Amazon File Details

```
complab3-7@complab3-7:~/experiment5$ head -5 ~/experiment5/Amazon.
OrderID,OrderDate,CustomerID,CustomerName,ProductID,ProductName,Ca
tity,UnitPrice,Discount,Tax,ShippingCost,TotalAmount,PaymentMethod
,State,Country,SellerID
ORD0000001,2023-01-31,CUST001504,Vihaan Sharma,P00014,Drone Mini,B
106.59,0.0,0.0,0.09,319.86,Debit Card,Delivered,Washington,DC,Indi
ORD0000002,2023-12-30,CUST000178,Pooja Kumar,P00040,Microphone,Hom
Style,1,251.37,0.05,19.1,1.74,259.64,Amazon Pay,Delivered,Fort Wor
es,SELL01298
ORD0000003,2022-05-10,CUST047516,Sneha Singh,P00044,Power Bank 200
banStyle,3,35.03,0.1,7.57,5.91,108.06,Debit Card,Delivered,Austin,
SELL00908
ORD0000004,2023-07-18,CUST030059,Vihaan Reddy,P00041,Webcam Full H
Zenith,5,33.58,0.15,11.42,5.53,159.66,Cash on Delivery,Delivered,C
,SELL01164
complab3-7@complab3-7:~/experiment5$ wc -l ~/experiment5/Amazon.cs
100001 /home/complab3-7/experiment5/Amazon.csv
complab3-7@complab3-7:~/experiment5$ du -h ~/experiment5/Amazon.cs
17M     /home/complab3-7/experiment5/Amazon.csv
complab3-7@complab3-7:~/experiment5$
```

Query 1 – Load Amazon Dataset

```
grunt> A = LOAD '/home/complab3-7/experiment5/Amazon.csv' USING PigStorage(',')
>>     AS (
>>         OrderID:chararray,
>>         OrderDate:chararray,
>>         CustomerID:chararray,
>>         CustomerName:chararray,
>>         ProductID:chararray,
>>         ProductName:chararray,
>>         Category:chararray,
>>         Brand:chararray,
>>         Quantity:int,
>>         UnitPrice:double,
>>         Discount:double,
>>         Tax:double,
>>         ShippingCost:double,
>>         TotalAmount:double,
>>         PaymentMethod:chararray,
>>         OrderStatus:chararray,
>>         City:chararray,
>>         State:chararray,
>>         Country:chararray,
>>         SellerID:chararray
>>     );
2026-08-24 15:51:11,605 [main] INFO  org.apache.hadoop.conf.Configuration.deprecati
on - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-per-checksum
grunt>
```

Query 2 – DESCRIBE the Dataset

```
grunt> DESCRIBE A;
A: {OrderID: chararray,OrderDate: chararray,CustomerID: chararray,C
rarray,ProductID: chararray,ProductName: chararray,Category: charar
rray,Quantity: int,UnitPrice: double,Discount: double,Tax: double,S
ble,TotalAmount: double,PaymentMethod: chararray,OrderStatus: chara
rray,State: chararray,Country: chararray,SellerID: chararray}
grunt>
```

```
is,IN,Canada,SELL01453)
grunt> B = GROUP A BY State;
grunt> DUMP B;
(ORD0096669,2020-02-15,CUST036849,Pooja Joshi,P00005,Laptop Sleeve,
218.95,0.1,9.85,2.65,209.56,Debit Card,Delivered,Columbus,OH,United
0)
(ORD0096670,2023-07-07,CUST030093,Aditya Sharma,P00003,Smartphone C
un,4,399.95,0.0,191.98,14.68,1806.46,Credit Card,Delivered,Jacksonv
tates,SELL01638)
(ORD0096671,2020-01-15,CUST010512,Simran Kapoor,P00030,Dress Shirt,
anStyle,1,434.58,0.05,33.03,7.62,453.5,Debit Card,Delivered,Philade
States,SELL00313)
(ORD0096672,2022-04-28,CUST004832,Neha Verma,P00027,Winter Jacket,B
,353.5,0.15,15.02,9.07,324.56,Amazon Pay,Delivered,Houston,TX,Unite
56)
(ORD0096673,2021-09-19,CUST025396,Vihaan Verma,P00039,Graphic Table
ors,Zenith,1,289.6,0.05,33.01,10.93,319.06,Credit Card,Delivered,Au
tates,SELL01720)
(ORD0096674,2020-01-12,CUST016202,Rohit Mehta,P00046,Car Charger,Ho
eTech,3,39.12,0.0,14.08,11.47,142.91,Net Banking,Delivered,Denver,C
SELL01699)
(ORD0096675,2020-01-01,CUST033002,Simran Kumar,P00014,Drone Mini,Cl
3,528.17,0.15,242.43,10.45,1599.71,Credit Card,Delivered,Denver,CO,
(ORD0096676,2020-03-17,CUST004312,Aman Reddy,P00027,Winter Jacket,H
nith,2,190.05,0.0,68.42,11.47,459.99,Credit Card,Delivered,Phoenix,
,SELL01429)
(ORD0096677,2021-06-25,CUST020002,Sneha Mehta,P00050,Novel Bestsell
Ease,5,291.97,0.0,72.99,8.55,1541.39,Net Banking,Delivered,Seattle,
,SELL00351)
(ORD0096678,2024-10-03,CUST026322,Aarav Kumar,P00007,Mechanical Key
chen,KiddoFun,5,301.18,0.0,120.47,4.62,1630.99,Credit Card,Returned
ralia,SELL01766)
(ORD0096679,2022-09-22,CUST039277,Simran Verma,P00011,Fitness Band,
s,ReadMore,4,64.89,0.15,26.48,3.49,250.6,Amazon Pay,Delivered,Seatt
00105)
(ORD0096680,2022-06-20,CUST043737,Simran Gupta,P00015,Instant Pot,B
4,270.32,0.0,0.0,6.53,1087.81,Cash on Delivery,Delivered,San Antoni
ELL00351)
(ORD0096681,2022-04-20,CUST033319,Vikas Singh,P00047,Memory Card 12
Apex,1,86.85,0.05,4.13,12.16,98.8,Credit Card,Delivered,Fort Worth,
,SELL00445)
(ORD0096682,2020-02-19,CUST029135,Vikas Kapoor,P00037,Router,Home &
```

Query 3 – GROUP the Dataset by State



```
ronics,Apex,1,528.33,0.1,85.59,7.73,568.82,Debit Card,Delivered,Seattle,WA,India,SE
LL00325),(ORD0034595,2024-12-05,CUST039179,Anjali Mehta,P00035,Desk Plant,Books,Bri
ghtLux,2,221.27,0.25,59.74,6.15,397.8,Credit Card,Delivered,Seattle,WA,United State
s,SELL01828),(ORD0034613,2020-02-09,CUST031841,Pooja Sharma,P00005,Laptop Sleeve,Ho
me & Kitchen,ReadMore,5,322.62,0.0,80.66,2.96,1696.72,Net Banking,Shipped,Seattle,W
A,India,SELL01720),(ORD0034653,2023-06-01,CUST034423,Vikas Mehta,P00048,Wireless Ch
arger,Electronics,BrightLux,5,193.33,0.0,0.0,3.28,969.93,UPI,Delivered,Seattle,WA,U
nited States,SELL00986),(ORD0034659,2020-04-18,CUST006114,Vihaan Sharma,P00011,Fitn
ess Band,Sports & Outdoors,BrightLux,3,592.56,0.0,88.88,9.78,1876.34,Debit Card,Del
ivered,Seattle,WA,United States,SELL00061),(ORD0034689,2024-08-08,CUST039978,Neha R
eddy,P00016,Air Fryer,Clothing,FitLife,2,105.02,0.0,16.8,4.65,231.49,Credit Card,De
livered,Seattle,WA,Canada,SELL01452),(ORD0001191,2020-10-03,CUST029058,Mohit Patel,
P00032,Board Game,Sports & Outdoors,UrbanStyle,5,480.11,0.0,120.03,11.05,2531.63,Cr
edit Card,Delivered,Seattle,WA,United States,SELL00760),(ORD0034705,2024-10-18,CUST
000807,Mohit Sharma,P00014,Drone Mini,Home & Kitchen,HomeEase,5,110.38,0.0,27.6,10.
51,590.01,Debit Card,Delivered,Seattle,WA,United States,SELL00223),(ORD0034727,2024
-12-17,CUST005782,Aman Reddy,P00042,Projector Mini,Books,FitLife,3,106.71,0.2,30.73
,5.52,292.35,Credit Card,Delivered,Seattle,WA,United States,SELL00657),(ORD0004760,
2024-08-27,CUST033537,Rohit Verma,P00020,Office Chair,Electronics,KiddoFun,4,586.96
,0.0,422.61,12.28,2782.73,Debit Card,Shipped,Seattle,WA,United States,SELL00553),(O
RD0034740,2021-06-13,CUST024638,Karan Verma,P00005,Laptop Sleeve,Home & Kitchen,Bri
ghtLux,4,526.84,0.1,94.83,12.11,2003.56,Credit Card,Shipped,Seattle,WA,United State
s,SELL00245),(ORD0034742,2020-03-03,CUST024945,Sunita Reddy,P00040,Microphone,Elect
ronics,UrbanStyle,3,323.05,0.0,48.46,4.53,1022.14,Amazon Pay,Delivered,Seattle,WA,U
nited States,SELL00014),(ORD0034762,2021-12-30,CUST027310,Rohit Gupta,P00013,Action
 Camera,Home & Kitchen,BrightLux,4,90.74,0.0,18.15,8.1,389.21,Credit Card,Shipped,S
eattle,WA,United States,SELL00743),(ORD0083634,2020-05-10,CUST011337,Anjali Joshi,P
00017,Electric Kettle,Clothing,Apex,1,416.41,0.0,49.97,5.41,471.79,UPI,Delivered,Se
attle,WA,United States,SELL01498),(ORD0004755,2022-12-03,CUST009779,Aditya Singh,P0
0001,Wireless Earbuds,Books,KiddoFun,3,470.24,0.0,169.29,10.19,1590.2,UPI,Delivered
,Seattle,WA,United States,SELL01587),(ORD0083641,2024-04-27,CUST020860,Priya Joshi,
P00043,HDMI Cable 2m,Electronics,HomeEase,4,194.4,0.0,38.88,7.51,823.99,UPI,Deliver
ed,Seattle,WA,India,SELL01388),(ORD0034818,2022-03-05,CUST000385,Sunita Sharma,P000
43,HDMI Cable 2m,Home & Kitchen,Apex,5,383.2,0.0,229.92,13.53,2159.45,Cash on Deliv
ery,Shipped,Seattle,WA,Australia,SELL01073),(ORD0083647,2024-01-28,CUST015582,Vivaa
n Gupta,P00014,Drone Mini,Toys & Games,CoreTech,1,368.68,0.0,66.36,10.29,445.33,Ama
zon Pay,Delivered,Seattle,WA,United States,SELL01729),(ORD0083666,2023-12-29,CUST04
4071,Sahil Sharma,P00041,Webcam Full HD,Clothing,Apex,5,165.42,0.0,66.17,6.31,899.5
8,Net Banking,Delivered,Seattle,WA,United States,SELL00242),(ORD0034858,2024-11-03,
CUST014237,Priya Joshi,P00027,Winter Jacket,Sports & Outdoors,UrbanStyle,4,296.08,0
.0,94.75,13.55,1292.62,Credit Card,Delivered,Seattle,WA,United States,SELL00054),(O
RD0034860,2024-08-27,CUST031302,Kabir Kumar,P00008,4K Monitor,Toys & Games,HomeEase
,1,73.03,0.0,0.0,0.33,73.36,Credit Card,Delivered,Seattle,WA,United States,SELL0011
```

```
.74,10.6,435.01,Credit Card,Pending,Seattle,WA,United States,SELL01275),(ORD003074
,2022-02-13,CUST011229,Sunita Kumar,P00022,Water Bottle,Sports & Outdoors,ReadMore
1,559.39,0.0,0.0,8.12,567.51,Amazon Pay,Delivered,Seattle,WA,United States,SELL010
3),(ORD0030790,2022-05-07,CUST038122,Rohit Reddy,P00026,Sunglasses,Books,HomeEase,
,60.12,0.0,12.02,14.38,266.88,Credit Card,Delivered,Seattle,WA,Australia,SELL01510
,(ORD0082787,2021-12-27,CUST010553,Vihaan Reddy,P00030,Dress Shirt,Clothing,Bright
ux,1,482.63,0.0,0.0,4.59,487.22,Amazon Pay,Delivered,Seattle,WA,Australia,SELL0101
),(ORD0030792,2021-04-20,CUST002294,Anjali Singh,P00016,Air Fryer,Home & Kitchen,K
ddoFun,5,408.92,0.0,163.57,14.71,2222.88,Debit Card,Delivered,Seattle,WA,United St
tes,SELL00655),(ORD0001260,2020-01-10,CUST003725,Vivaan Reddy,P00039,Graphic Table
,Sports & Outdoors,FitLife,5,324.69,0.0,292.22,11.9,1927.57,Amazon Pay,Delivered,S
attle,WA,India,SELL00388),(ORD0030817,2021-09-11,CUST024836,Rohit Sharma,P00020,Of
ice Chair,Toys & Games,CoreTech,5,496.66,0.0,0.0,0.62,2483.92,Debit Card,Delivered
Seattle,WA,United Kingdom,SELL00044),(ORD0030824,2024-09-20,CUST002528,Arjun Singh
P00035,Desk Plant,Books,Zenith,3,24.49,0.0,3.67,2.42,79.56,Credit Card,Delivered,S
attle,WA,Canada,SELL00426),(ORD0095382,2024-06-18,CUST015195,Vikas Sharma,P00046,C
r Charger,Books,HomeEase,1,587.94,0.2,37.63,13.71,521.69,Credit Card,Pending,Seatt
e,WA,United States,SELL01816),(ORD0030830,2021-09-01,CUST004097,Sahil Kapoor,P0004
,Microphone,Home & Kitchen,HomeEase,3,366.16,0.1,118.64,14.27,1121.54,Credit Card,
elivered,Seattle,WA,United States,SELL01335),(ORD0030842,2020-05-17,CUST046766,Sun
ta Sharma,P00035,Desk Plant,Sports & Outdoors,BrightLux,1,250.07,0.0,30.01,2.96,28
.04,Credit Card,Pending,Seattle,WA,United States,SELL00146),(ORD0030855,2023-08-23
CUST041056,Aditya Gupta,P00020,Office Chair,Sports & Outdoors,Zenith,4,356.11,0.05
108.26,1.81,1463.29,Net Banking,Delivered,Seattle,WA,United States,SELL01247),(ORD
082840,2022-05-16,CUST041666,Kabir Verma,P00007,Mechanical Keyboard,Sports & Outdo
rs,NexPro,2,112.94,0.05,10.73,9.79,235.11,Credit Card,Delivered,Seattle,WA,Canada,
ELL00247),(ORD0030884,2020-04-08,CUST027576,Pooja Gupta,P00040,Microphone,Toys & G
mes,Apex,4,344.22,0.0,110.15,13.44,1500.47,Amazon Pay,Delivered,Seattle,WA,United
tates,SELL00055),(ORD0001257,2024-08-23,CUST003265,Neha Kumar,P00005,Laptop Sleeve
Sports & Outdoors,Apex,3,71.65,0.05,10.21,9.65,224.06,UPI,Delivered,Seattle,WA,Uni
ed States,SELL00779),(ORD0030946,2021-11-19,CUST001317,Aman Reddy,P00034,Desk Orga
izer,Home & Kitchen,CoreTech,5,89.25,0.1,72.29,1.97,475.89,Credit Card,Shipped,Sea
tle,WA,United States,SELL01593),(ORD0099770,2023-06-21,CUST026112,Aarav Verma,P000
9,Children's Book,Sports & Outdoors,NexPro,2,412.97,0.0,99.11,8.83,933.88,Credit C
rd,Shipped,Seattle,WA,United States,SELL01169),(ORD0030956,2024-02-01,CUST040611,A
rav Gupta,P00044,Power Bank 20000mAh,Toys & Games,Zenith,2,167.45,0.3,18.75,10.34,
63.52,Credit Card,Pending,Seattle,WA,United Kingdom,SELL00474),(ORD0031012,2024-04
22,CUST032121,Rohit Mehta,P00012,Noise Cancelling Headphones,Electronics,UrbanStyl
,5,206.66,0.0,51.66,0.99,1085.95,Cash on Delivery,Delivered,Seattle,WA,United Stat
s,SELL01674)})
(State,{(OrderID,OrderDate,CustomerID,CustomerName,ProductID,ProductName,Category,
rand,,,,,,,PaymentMethod,OrderStatus,City,State,Country,SellerID)})
grunt>
```

Query 4 – Count Orders for Each State

```
grunt> C = FOREACH B GENERATE group AS State, COUNT(A) AS OrderCount;
grunt> DUMP C;
```

```
Total records proactively spilled: 0

Job DAG:
job_local361647742_0004


2026-08-24 15:55:17,471 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:55:17,472 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:55:17,472 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:55:17,473 [main] WARN  org.apache.pig.backend.hadoop.
mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCAR
ION_FAILED 6 time(s).
2026-08-24 15:55:17,473 [main] INFO  org.apache.pig.backend.hadoop.
mapReduceLayer.MapReduceLauncher - Success!
2026-08-24 15:55:17,473 [main] INFO  org.apache.hadoop.conf.Configu
on - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-pe
2026-08-24 15:55:17,473 [main] WARN  org.apache.pig.data.SchemaTupl
aTupleBackend has already been initialized
2026-08-24 15:55:17,478 [main] INFO  org.apache.hadoop.mapreduce.li
tFormat - Total input files to process : 1
2026-08-24 15:55:17,478 [main] INFO  org.apache.pig.backend.hadoop.
util.MapRedUtil - Total input paths to process : 1
(AZ,4941)
(CA,19921)
(CO,4991)
(DC,5021)
(FL,5107)
(IL,5020)
(IN,4970)
(NC,5110)
(NY,4949)
(OH,5021)
(PA,5014)
(TX,24896)
(WA,5039)
(State,1)
grunt>
```

Query 5 – Calculate Total Sales for Each State

```
grunt> D = GROUP A BY State;
grunt> E = FOREACH D GENERATE group AS State, SUM(A.TotalAmount) AS Total
grunt> DUMP E;
```

```
Total records proactively spilled: 0

Job DAG:
job_local585734000_0005


2026-08-24 15:57:01,868 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:57:01,869 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:57:01,869 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:57:01,870 [main] WARN  org.apache.pig.backend.hadoop.
mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCAR
ION_FAILED 6 time(s).
2026-08-24 15:57:01,870 [main] INFO  org.apache.pig.backend.hadoop.
mapReduceLayer.MapReduceLauncher - Success!
2026-08-24 15:57:01,870 [main] INFO  org.apache.hadoop.conf.Configu
on - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-pe
2026-08-24 15:57:01,870 [main] WARN  org.apache.pig.data.SchemaTupl
aTupleBackend has already been initialized
2026-08-24 15:57:01,875 [main] INFO  org.apache.hadoop.mapreduce.li
tFormat - Total input files to process : 1
2026-08-24 15:57:01,875 [main] INFO  org.apache.pig.backend.hadoop.
util.MapRedUtil - Total input paths to process : 1
(AZ,4504324.190000006)
(CA,1.823119679000001E7)
(CO,4638498.97)
(DC,4520876.110000017)
(FL,4597177.27000001)
(IL,4632847.440000005)
(IN,4609048.430000001)
(NC,4747701.729999998)
(NY,4554358.079999998)
(OH,4615442.3200000115)
(PA,4650673.879999988)
(TX,2.2862540539999936E7)
(WA,4660962.17000001)
(State,)
grunt>
```

Query 5 – Calculate Total Sales for Each State

```
(State,1)
grunt> D = GROUP A BY State;
grunt> E = FOREACH D GENERATE group AS State, SUM(A.TotalAmount) AS Total
grunt> DUMP E;
```

```
Total records proactively spilled: 0

Job DAG:
job_local585734000_0005


2026-08-24 15:57:01,868 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:57:01,869 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:57:01,869 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:57:01,870 [main] WARN  org.apache.pig.backend.hadoop.
mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCAR
ION_FAILED 6 time(s).
2026-08-24 15:57:01,870 [main] INFO  org.apache.pig.backend.hadoop.
mapReduceLayer.MapReduceLauncher - Success!
2026-08-24 15:57:01,870 [main] INFO  org.apache.hadoop.conf.Configu
on - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-pe
2026-08-24 15:57:01,870 [main] WARN  org.apache.pig.data.SchemaTupl
aTupleBackend has already been initialized
2026-08-24 15:57:01,875 [main] INFO  org.apache.hadoop.mapreduce.li
tFormat - Total input files to process : 1
2026-08-24 15:57:01,875 [main] INFO  org.apache.pig.backend.hadoop.
util.MapRedUtil - Total input paths to process : 1
(AZ,4504324.190000006)
(CA,1.823119679000001E7)
(CO,4638498.97)
(DC,4520876.110000017)
(FL,4597177.27000001)
(IL,4632847.440000005)
(IN,4609048.430000001)
(NC,4747701.729999998)
(NY,4554358.079999998)
(OH,4615442.3200000115)
(PA,4650673.879999988)
(TX,2.2862540539999936E7)
(WA,4660962.17000001)
(State,)
grunt>
```

Query 6 – Calculate Average Order Amount

```
grunt> F = FOREACH D GENERATE group AS State, AVG(A.TotalAmount) A
unt;
grunt> DUMP F;
```

```
Total records proactively spilled: 0

Job DAG:
job_local1424068078_0006


2026-08-24 15:57:48,792 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:57:48,792 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:57:48,793 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:57:48,794 [main] WARN  org.apache.pig.backend.hadoop.
mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCAR
ION_FAILED 6 time(s).
2026-08-24 15:57:48,794 [main] INFO  org.apache.pig.backend.hadoop.
mapReduceLayer.MapReduceLauncher - Success!
2026-08-24 15:57:48,795 [main] INFO  org.apache.hadoop.conf.Configu
on - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-pe
2026-08-24 15:57:48,795 [main] WARN  org.apache.pig.data.SchemaTupl
aTupleBackend has already been initialized
2026-08-24 15:57:48,800 [main] INFO  org.apache.hadoop.mapreduce.li
tFormat - Total input files to process : 1
2026-08-24 15:57:48,800 [main] INFO  org.apache.pig.backend.hadoop.
util.MapRedUtil - Total input paths to process : 1
(AZ,911.621977332525)
(CA,915.1747798805286)
(CO,929.3726647966339)
(DC,900.3935690101607)
(FL,900.1717779518327)
(IL,922.8779760956186)
(IN,927.3739295774649)
(NC,929.1001428571424)
(NY,920.2582501515454)
(OH,919.2277076279648)
(PA,927.5376705225344)
(TX,918.3218404562956)
(WA,924.9776086525125)
(State,)
grunt>
```

Storing State-wise Total Sales Result

```
Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime
me      MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReduc
as      Feature Outputs
job_local397324360_0007 1       1       n/a     n/a     n/a     n/a
n/a     n/a     A,D,E   GROUP_BY,COMBINER       /home/complab3-7/ex
total_sales,

Input(s):
Successfully read 100001 records from: "/home/complab3-7/experiment

Output(s):
Successfully stored 14 records in: "/home/complab3-7/experiment5/st

Counters:
Total records written : 14
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local397324360_0007


2026-08-24 15:58:23,389 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:58:23,389 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:58:23,390 [main] INFO  org.apache.hadoop.metrics.jvm.
not initialize JVM Metrics with processName=JobTracker, sessionId=
lized
2026-08-24 15:58:23,390 [main] WARN  org.apache.pig.backend.hadoop.
mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCAR
ION_FAILED 6 time(s).
2026-08-24 15:58:23,390 [main] INFO  org.apache.pig.backend.hadoop.
mapReduceLayer.MapReduceLauncher - Success!
grunt>
```

Verifying Stored Pig Output

```
complab3-7@complab3-7:~/experiment5$ ls -lh ~/experiment5/state_total_s
total 4.0K
-rw-r--r-- 1 complab3-7 complab3-7 277 Aug 24 15:58 part-r-00000
-rw-r--r-- 1 complab3-7 complab3-7   0 Aug 24 15:58 _SUCCESS
complab3-7@complab3-7:~/experiment5$
```

View the Stored Result

```
complab3-7@complab3-7:~/experiment5$ cat ~/experiment5/state_total_sales/part-m-00
00
cat: /home/complab3-7/experiment5/state_total_sales/part-m-00000: No such file or
irectory
complab3-7@complab3-7:~/experiment5$
```

Conclusion:	Thus students were able to successfully install Apache PIG and apply various operators and commands to manage the big data.
References:	Put the reference of resources used to perform this experiment Moodle E learn :- https://elearn.dbit.in/mod/resource/view.php?id=18696 https://elearn.dbit.in/mod/resource/view.php?id=18695

Lab Manual - BDA

Experiment No: 6

Name: Navya Nair
Roll No: 24
Div B
Batch B
Performance Date : 29/9/26

Topic: Implement Bloom Filter using any programming language. (for specific Application)

Prerequisite: - Familiarity with the programming language

- Basic concept of Bloom Filter

Mapping With COs: CSL702.5

Objective: Implement and apply Bloom filters for any one appropriate real world application.

Outcomes: Students will be able to understand the concept of Bloom filter and its usage and also be able to implement it for any real world problem.

Instructions: This experiment is a compulsory experiment. All the students are required to perform this experiment in a group.

Deliverables: **Submission on Moodle:**

Take snapshot of following task:
Explain one Real World Problem for which you want to create a Bloom Filter.

State: Its Input and Expected Outputs

Bloom Filter [Code]Results: Input and Output

- Demo during Practical

1. Real World Problem

In a college, thousands of students may be registered for courses, examinations, or university activities. Whenever a Student ID needs to be checked, searching the complete student database repeatedly can take additional time and resources.A Bloom Filter can be used as a fast preliminary checker to determine whether a Student ID is already registered.The Bloom Filter does not store the complete Student ID. Instead, it uses a bit array and multiple hash functions to represent the registered IDs.

If the Bloom Filter says:

- Definitely Not Registered → the Student ID is definitely not present.
- Possibly Registered → the Student ID may be present, so the actual database can be checked.

This is useful because Bloom Filters provide fast membership checking while using relatively little memory.

2. Input and Expected Output

Input : The system takes two types of input

1. Student IDs to be checked

Expected Output : For every queried Student ID, the system produces a Bloom Filter result

Bloom Filter Result :

1. Not Registered : ID is definitely absent
2. Possibly Registered : ID may be present and should be verified

Bloom Filter [Code] :

```
import math
import hashlib
```

```
import random

class StudentIDBloomFilter:
    def _init_(self, expected_elements: int, false_positive_rate: float):
        self.n = expected_elements
        self.p = false_positive_rate

        # Calculate optimal bit array size: m = -(n * ln(p)) / (ln(2)^2)
        self.size = max(1, int(- (self.n * math.log(self.p)) / (math.log(2) ** 2)))

        # Calculate optimal number of hash functions: k = (m / n) * ln(2)
        self.hash_count = max(1, round((self.size / self.n) * math.log(2)))

        # Bit array representation
        self.bit_array = [0] * self.size

        # Simulated database storing actual records to verify ground truth
        self.mock_database = set()

    def _get_hashes(self, item: str):
        """Generates k hash indices using the Kirsch-Mitzenmacher double-hashing technique."""
        clean_item = item.strip().upper()
        h1 = int(hashlib.md5(clean_item.encode('utf-8')).hexdigest(), 16)
        h2 = int(hashlib.sha256(clean_item.encode('utf-8')).hexdigest(), 16)

        indices = []
        for i in range(self.hash_count):
            index = (h1 + i * h2) % self.size
            indices.append(index)
        return indices

    def register_id(self, student_id: str) -> bool:
        """Attempts to register a student ID."""
        clean_id = student_id.strip().upper()

        # Step 1: Query Bloom Filter
        if self.is_possibly_registered(clean_id):
            # Step 2: Query simulated database to handle collisions / true positives
            if clean_id in self.mock_database:
                return False  # Already registered in DB
            else:
                print("  [!] Notice: False Positive occurred! Filter returned positive, but ID is not in DB.")
```

```
        # Set bits in Bloom Filter
        for idx in self._get_hashes(clean_id):
            self.bit_array[idx] = 1

        # Add to underlying database
        self.mock_database.add(clean_id)
        return True

    def is_possibly_registered(self, student_id: str) -> bool:
        """Returns False if definitely absent, True if possibly present."""
        clean_id = student_id.strip().upper()
        for idx in self._get_hashes(clean_id):
            if self.bit_array[idx] == 0:
                return False
        return True


def generate_initial_students():
    """Generates 10 student IDs: 3 mandatory + 7 randomly selected from range 2023120000 to
2023120100."""
    mandatory_ids = ["2023120045", "2023120032", "2023120047"]
    all_possible = [f"202312{i:04d}" for i in range(0, 101)]
    pool = [sid for sid in all_possible if sid not in mandatory_ids]

    random_sample = random.sample(pool, 7)
    initial_ids = mandatory_ids + random_sample
    initial_ids.sort()
    return initial_ids


def main():
    print("=" * 60)
    print("   STUDENT ID REGISTRATION CHECKER (BLOOM FILTER)")
    print("=" * 60)

    # Initialize Filter: capacity = 100, false positive rate = 5%
    capacity = 100
    fp_rate = 0.05
    bf = StudentIDBloomFilter(capacity, fp_rate)

    # Silently pre-populate the 10 student IDs
```

```
initial_students = generate_initial_students()
for sid in initial_students:
    bf.register_id(sid)

# Interactive Loop
while True:
    print("\n--- MENU ---")
    print("1. Register a new Student ID")
    print("2. Check if a Student ID exists (Query)")
    print("3. View All Registered Database IDs")
    print("4. Inspect Raw Bit Array (First 64 bits)")
    print("5. Exit")

    choice = input("Enter choice (1-5): ").strip()

    if choice == "1":
        sid = input("Enter Student ID to register (e.g., 2023120080): ").strip()
        if not sid:
            print("[!] Error: ID cannot be empty.")
            continue

        success = bf.register_id(sid)
        if success:
            print(f"[SUCCESS] Student ID '{sid}' registered and bit array updated.")
        else:
            print(f"[REJECTED] Student ID '{sid}' is ALREADY REGISTERED in the system.")

    elif choice == "2":
        sid = input("Enter Student ID to check: ").strip()
        if not sid:
            continue

        hash_indices = bf._get_hashes(sid)
        print(f"  -> Generated Hash Indices: {hash_indices}")

        if not bf.is_possibly_registered(sid):
            print(f"[RESULT] '{sid}' is DEFINITELY NOT REGISTERED.")
            print("         Reason: At least one hash index contains bit 0.")
            print("         Action: Disk lookup avoided. Can proceed to registration.")
        else:
            print(f"[RESULT] '{sid}' is POSSIBLY REGISTERED.")
            print("         Reason: All corresponding bit positions are set to 1.")
```

```
            if sid in bf.mock_database:
                print("        Database Verification: MATCH CONFIRMED (Actual record exists in
DB).")
            else:
                print("        Database Verification: FALSE POSITIVE (Collision occurred; not in
DB).")

        elif choice == "3":
            print("\nCurrently Registered IDs in Database:")
            for idx, sid in enumerate(sorted(bf.mock_database), 1):
                print(f" {idx:2d}. {sid}")

        elif choice == "4":
            preview_bits = "".join(str(b) for b in bf.bit_array[:64])
            print("\nBit Array Preview (first 64 bits):")
            print(f"[{preview_bits}]...")

        elif choice == "5":
            print("\nExiting practical demo. Goodbye!")
            break
        else:
            print("[!] Invalid choice. Please select 1 to 5.")


if _name___== "_main_":
    main()
```

Results: Input and Output

```
============================================================
   STUDENT ID REGISTRATION CHECKER (BLOOM FILTER)
============================================================

--- MENU ---
1. Register a new Student ID
2. Check if a Student ID exists (Query)
3. View All Registered Database IDs
4. Inspect Raw Bit Array (First 64 bits)
5. Exit
Enter choice (1-5):  1
Enter Student ID to register (e.g., 2023120080):  2023120041
[SUCCESS] Student ID '2023120041' registered and bit array updated.

--- MENU ---
1. Register a new Student ID
2. Check if a Student ID exists (Query)
3. View All Registered Database IDs
4. Inspect Raw Bit Array (First 64 bits)
5. Exit
Enter choice (1-5):  2
Enter Student ID to check:  2023120045
  -> Generated Hash Indices: [219, 186, 153, 120]
[RESULT] '2023120045' is POSSIBLY REGISTERED.
         Reason: All corresponding bit positions are set to 1.
         Database Verification: MATCH CONFIRMED (Actual record exists in DB).


 --- MENU ---
 1. Register a new Student ID
 2. Check if a Student ID exists (Query)
 3. View All Registered Database IDs
 4. Inspect Raw Bit Array (First 64 bits)
 5. Exit
 Enter choice (1-5):  2
 Enter Student ID to check:  2023120132
   -> Generated Hash Indices: [176, 128, 80, 32]
 [RESULT] '2023120132' is DEFINITELY NOT REGISTERED.
          Reason: At least one hash index contains bit 0.
          Action: Disk lookup avoided. Can proceed to registration.

 --- MENU ---
 1. Register a new Student ID
 2. Check if a Student ID exists (Query)
 3. View All Registered Database IDs
 4. Inspect Raw Bit Array (First 64 bits)
 5. Exit
 Enter choice (1-5):  3

 Currently Registered IDs in Database:
   1. 2023120015
   2. 2023120020
   3. 2023120031
   4. 2023120032
   5. 2023120033
   6. 2023120041
   7. 2023120043
   8. 2023120045
   9. 2023120047
  10. 2023120075
  11. 2023120097
```

```
--- MENU ---
1. Register a new Student ID
2. Check if a Student ID exists (Query)
3. View All Registered Database IDs
4. Inspect Raw Bit Array (First 64 bits)
5. Exit
Enter choice (1-5):  4

Bit Array Preview (first 64 bits):
[0000000010000000000010000000000010000000000000000000000000000000]...

--- MENU ---
1. Register a new Student ID
2. Check if a Student ID exists (Query)
3. View All Registered Database IDs
4. Inspect Raw Bit Array (First 64 bits)
5. Exit
Enter choice (1-5):  5

Exiting practical demo. Goodbye!
```

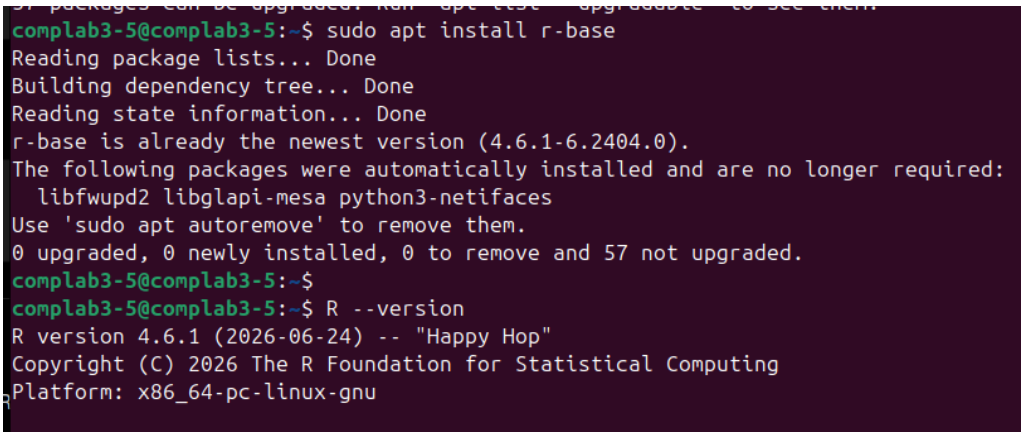
Conclusion: Students will be able to implement Bloom filters for real world problems.
References: https://dl.acm.org/doi/10.1145/362686.362692

Put the reference of resources used to perform this experiment.

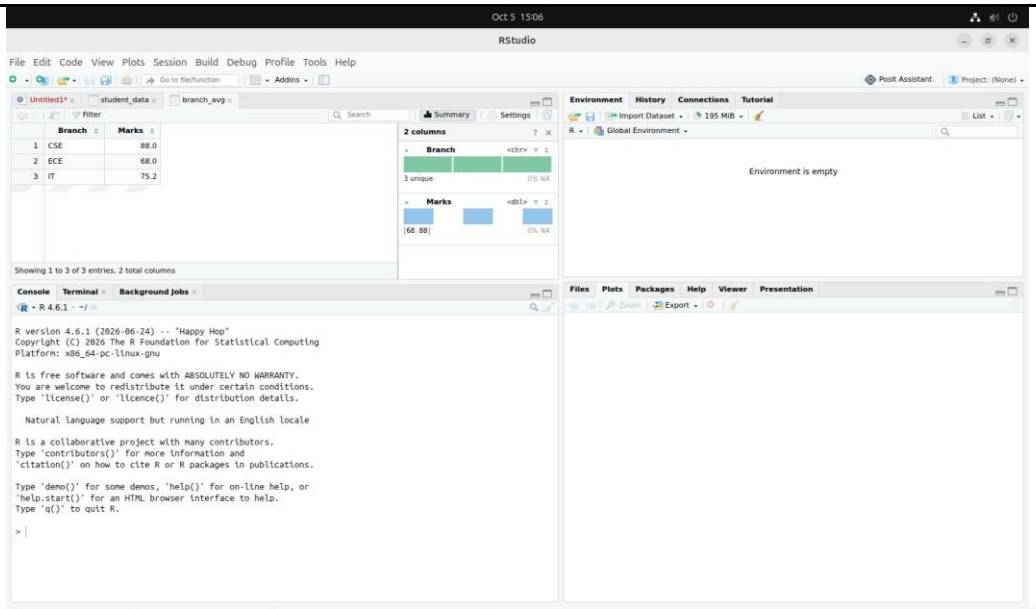
Experiment No: 7

Name: Navya Nair Roll No. : 24

Batch: B Performance Date :8/10/26

Topic:	Data Analysis & Data Visualization using Hive/PIG/R/Tableau.
Prerequisite:	Basic knowledge of Data Analysis & data visualization tools like Hive/PIG/R/Tabl
Mapping With COs:	CSL702.4
Objective:	Able to be used for data analysis, statistical modeling, and data visualization tasks.
Outcomes:	Analyze the datasets using Data Analysis tools & generate reports using data visualization to capabilities.
Instructions:	This experiment is a compulsory experiment. All the students are required to perform this ex individually.
Deliverables:	**Submission:** 1. Installation steps snapshots. 

```
complab3-5@complab3-5:~$ sudo apt install r-base
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
r-base is already the newest version (4.6.1-6.2404.0).
The following packages were automatically installed and are no longer required:
  libfwupd2 libglapi-mesa python3-netifaces
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 57 not upgraded.
complab3-5@complab3-5:~$
complab3-5@complab3-5:~$ R --version
R version 4.6.1 (2026-06-24) -- "Happy Hop"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu
```




```
Console  Terminal  Background Jobs
R 4.6.1 · ~/

> --version
Error in -version : invalid argument to unary operator

> version
               _
platform       x86_64-pc-linux-gnu
arch           x86_64
os             linux-gnu
system         x86_64, linux-gnu
status
major          4
minor          6.1
year           2026
month          06
day            24
svn rev        90187
language       R
version.string R version 4.6.1 (2026-06-24)
nickname       Happy Hop
> print("R is working successfully")
[1] "R is working successfully"
>
```

2. Detail about Datasets

The dataset contains information about students and their academic performance.

Column	Description	Data Type
Student_ID	Unique student identifier	Integer
Gender	Gender of student	Character
Study_Hours	Daily study hours	Numeric
Attendance	Attendance percentage	Numeric
Assignment_Score	Assignment score	Numeric
Internal_Marks	Internal examination marks	Numeric
Final_Marks	Final examination marks	Numeric

Student_ID	Gender	Study_Hours	Attendance	Assignment_Score	Internal_Marks	Final_Marks
1	Male	2	72	65	60	62
2	Female	4	85	78	82	84
3	Male	3	80	72	70	73
4	Female	5	92	88	90	91
5	Male	1	65	55	52	50
6	Female	3	78	75	73	76
7	Male	4	88	82	80	83
8	Female	2	70	68	64	65
9	Male	5	95	91	92	94
10	Female	4	86	85	84	86
11	Male	3	76	70	68	71
12	Female	5	94	90	91	93
13	Male	2	68	60	58	59
14	Female	3	82	77	75	78
15	Male	4	89	84	81	85

3. Data Analysis

1. Display structure of dataset

```
> str(student_data)
'data.frame':   15 obs. of  7 variables:
 $ Student_ID      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Gender          : chr  "Male" "Female" "Male" "Female" ...
 $ Study_Hours     : num  2 4 3 5 1 3 4 2 5 4 ...
 $ Attendance      : num  72 85 80 92 65 78 88 70 95 86 ...
 $ Assignment_Score: num  65 78 72 88 55 75 82 68 91 85 ...
 $ Internal_Marks  : num  60 82 70 90 52 73 80 64 92 84 ...
 $ Final_Marks     : num  62 84 73 91 50 76 83 65 94 86 ...
```

2. Summary of dataset

```
> summary(student_data)
   Student_ID        Gender    Study_Hours      Attendance    Assignment_Score Internal_Marks   Final_Marks
 Min.   : 1.0   Length   :15   Min.   :1.000   Min.   :65.00   Min.   :55.0     Min.   :52.00   Min.   :50.00
 1st Qu.: 4.5   N.unique : 2   1st Qu.:2.500   1st Qu.:74.00   1st Qu.:69.0     1st Qu.:66.00   1st Qu.:68.00
 Median : 8.0   N.blank  : 0   Median :3.000   Median :82.00   Median :77.0     Median :75.00   Median :78.00
 Mean   : 8.0   Min.nchar: 4   Mean   :3.333   Mean   :81.33   Mean   :76.0     Mean   :74.67   Mean   :76.67
 3rd Qu.:11.5   Max.nchar: 6   3rd Qu.:4.000   3rd Qu.:88.50   3rd Qu.:84.5     3rd Qu.:83.00   3rd Qu.:85.50
 Max.   :15.0                  Max.   :5.000   Max.   :95.00   Max.   :91.0     Max.   :92.00   Max.   :94.00
```

3. Mean, Median and Standard deviation

```
> mean(student_data$Final_Marks)
[1] 76.66667
> median(student_data$Final_Marks)
[1] 78
```

```
> sd(student_data$Final_Marks)
[1] 13.21615
```

4. Minimum and Maximum

```
> max(student_data$Final_Marks)
[1] 94
> min(student_data$Final_Marks)
[1] 50
```

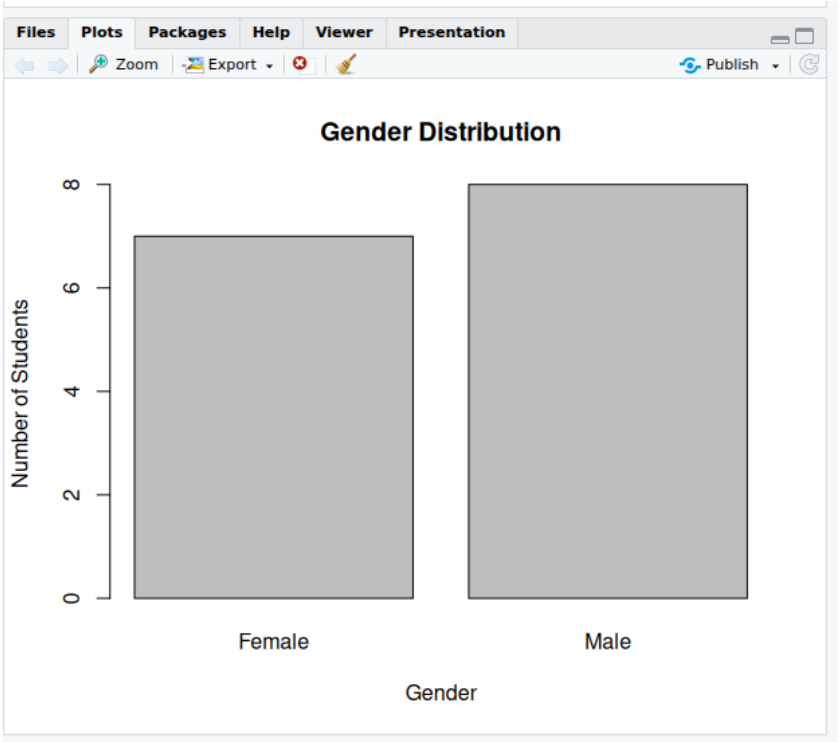
5. Gender count

```
> table(student_data$Gender)

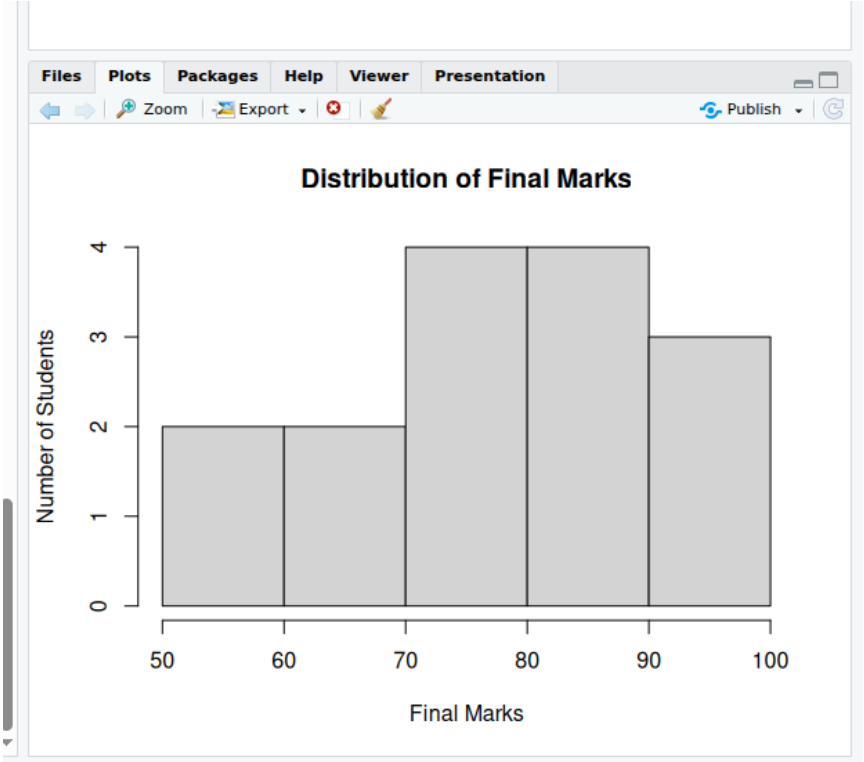
Female   Male
     7      8
```

4. Data Visualization

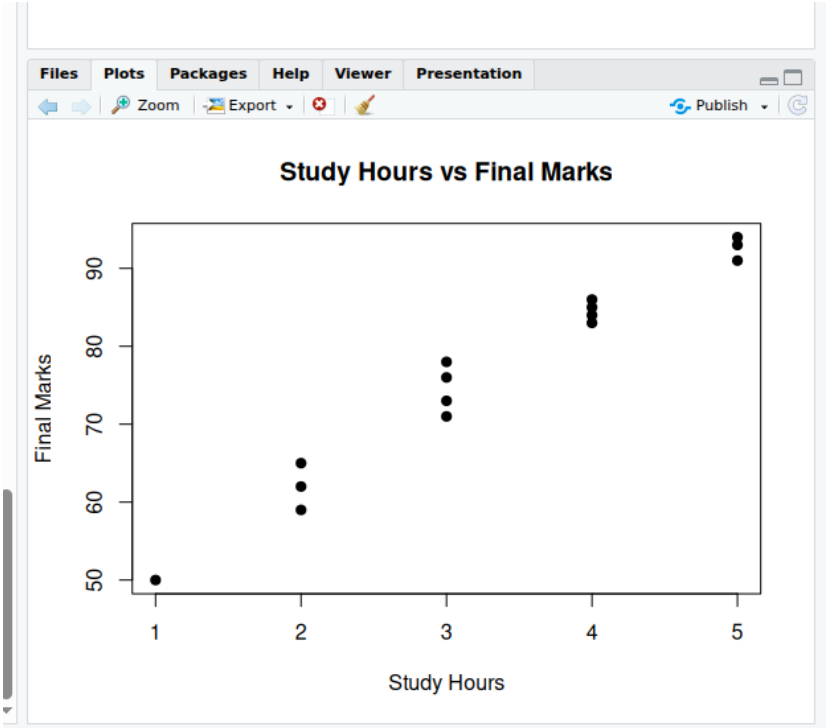
1. Bar Chart

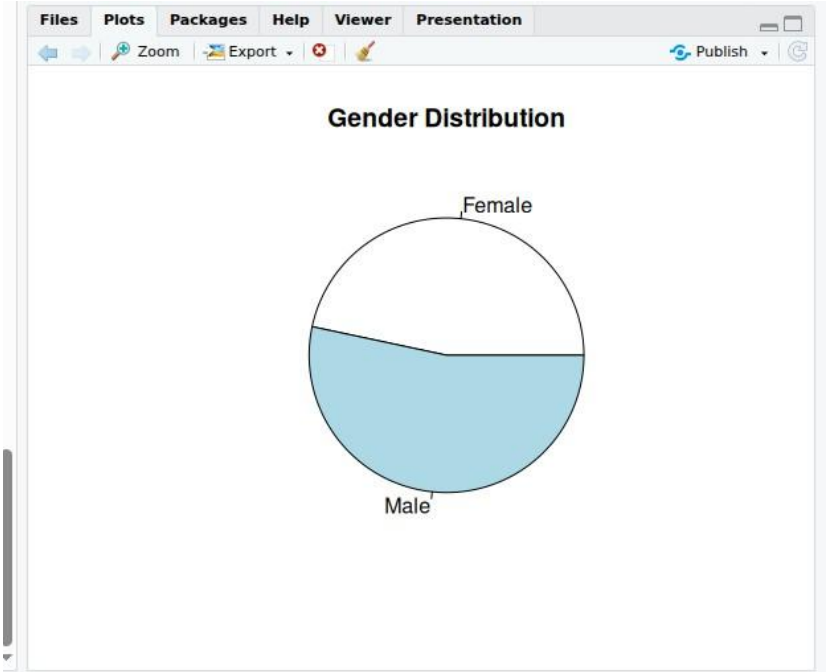


2. Histogram



3. Scatter plot



	4.Pie Chart 
Conclusion:	Students will be able to successfully perform Data Analysis & Data Visualization us Hive/PIG/R/Tableau.
References:	Put the reference of resources used to perform this experiment. R installation on windows https://teacherscollege.screenstepslive.com/a/1108074-install-r-and windows R installation on ubuntu https://www.geeksforgeeks.org/r-language/how-to-install-r-studio-on-windo ws-and-linux/ R basics https://www.geeksforgeeks.org/r-language/r-tutorial/ R -Data visualisation https://www.geeksforgeeks.org/r-language/data-visualization-in-r/ R-data Analysis https://www.geeksforgeeks.org/r-language/data-analysis-using-r/

Experiment No: 8

Name: Navya Nair **Roll No. : 24**

Batch: B **Performance Date :9/10/26**

Topic:	Mini Project: One real life large data application to be implemented (Use standard Datasets available on the web).
Prerequisite:	Knowledge of Big data analytics algorithms
Mapping With COs:	**CSL702.6**
Objective:	Able to apply big data technology and algorithms to solve real life applications.
Outcomes:	Students will be able to implement real life application using big data technology.
Instructions:	This experiment is a compulsory experiment. All the students are required to perform this experiment individually.
Deliverables:	**Submission:** **Create group of 3 people.Each member will upload the project.** Description of large dataset 1. Problem Definition of any real life application (For example: 1.Streaming data analysis ,use flume for data capture 2. HIVE/PYSpark for analysis of twitter data, chat data, weblog analysis etc. 3.Recommendation System (for example: Health Care System, Stock Market Prediction, Movie Recommendation, etc.) 4.SpatioTemporal DataAnalytics) 2. Explain big data technique/algorithm used with logic. 3. Take screenshots of your input,code and output screens.Explain the logic and interpretation of results. 4. Prepare a presentation of 5-7 minutes explaining the mini project-introduction to topic,input,use of big data algorithm,visualization and interpretation of result with demo of mini project. **This presentation each team has to give only during respective BDA lab between 5-10th oct 2026.**
Mini Project Report	**1. Introduction to Topic** **1.1 Problem definition** When a customer places an online order, the platform shows an estimated delivery date. Some orders arrive later than that date, which hurts customer trust and makes logistics harder to plan. SupplyChainX studies late deliveries in the public Olist Brazilian E-Commerce dataset using Apache Spark. The project has two aims: to measure when and where delays are more common, and to test whether order information can help identify late-delivery risk. **1.2 Objective** • Validate, clean and join several connected CSV files in PySpark. • Define a late-delivery label and compare delivery performance by month, seller and seller-state-to-customer-state route with Spark SQL. • Explore the results with interactive Plotly charts and store the processed tables as Parquet. • Flag unusually short or long delivery durations with the IQR rule.

- Compare Logistic Regression, Random Forest and Gradient-Boosted Trees in Spark MLlib using precision, recall, F1, ROC-AUC and confusion counts.

1.3 Why big data tools

The logistics information is spread over several connected tables that must be combined before any analysis. Spark provides DataFrames and SQL for joining and aggregating such tables, a machine learning library (MLlib) for model comparison, and works with columnar storage such as Parquet. The same code would apply to a much larger dataset. Here Spark runs locally in a Google Colab runtime, so the project demonstrates Spark processing but is not a multi-machine cluster.

1.4 Tools and technology

Component	Role in SupplyChainX
Apache Spark (PySpark)	Reading, validating, cleaning and joining the CSV files
Spark SQL	Delivery performance by month, seller and route
Spark MLlib	Logistic Regression, Random Forest, Gradient-Boosted Trees
Parquet	Columnar storage of processed tables
Plotly	Interactive charts for exploration
Google Colab	Notebook environment, with Spark in local mode

2. Dataset Overview

2.1 Source

The data is the public Olist Brazilian E-Commerce dataset. It contains around 100,000 orders placed between 2016 and 2018, supplied as several connected CSV files.

2.2 Tables used

Table	Content	Link to other tables
orders	Order records with purchase, delivered and estimated delivery dates	customer_id to customers; order_id to order items
order items	Items in each order	product_id to products; seller_id to sellers
products	Product details	product_id
customers	Customer details including state	customer_id
sellers	Seller details including state	seller_id

2.3 Validation and preparation

- The expected columns of each file are validated before use.
- Date columns and numeric fields are cleaned.
- The files are joined in PySpark into one analysis table.

2.4 Late-delivery label

A delivery is late when its delivered date is after its estimated date. Same-day and early deliveries count as on time. Orders without the necessary dates are left unlabeled and are not used for training or evaluation.

2.5 Avoiding target leakage

For prediction, the actual delivery date and other outcome fields are excluded from the predictors. Otherwise the model would see the answer it is supposed to predict.

2.6 Key figures

Item	Value
Orders	around 100,000
Period	2016 to 2018

3. Dataset Visualisation

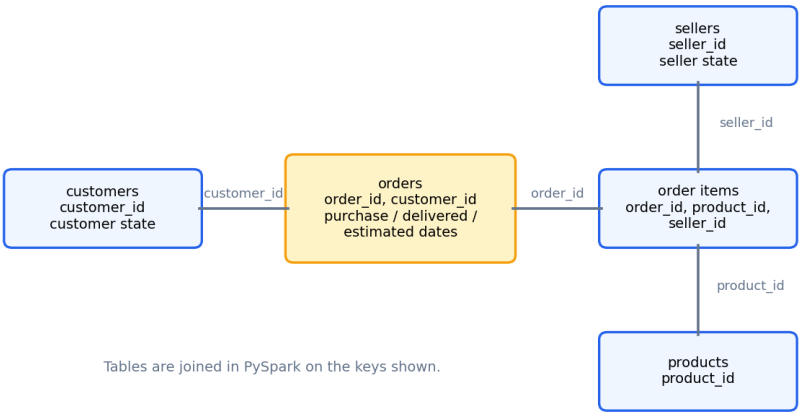


Figure 1. The connected Olist tables and the keys used to join them in PySpark.

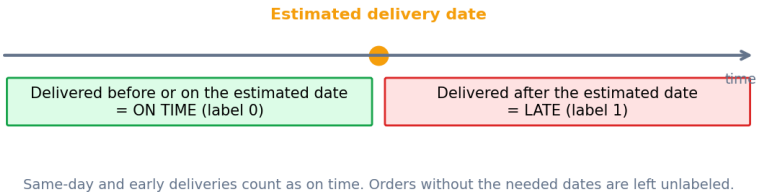


Figure 2. How the late-delivery label is defined.

The notebook explores the processed data with interactive Plotly charts. Insert screenshots of these charts below.

Chart	What it shows
Order volume	Number of orders over time
Late rate	Share of late deliveries by month, seller and route
Delivery-time distribution	How long deliveries take, and where unusual durations lie
Seller comparison	Late rates of sellers compared with each other
Route patterns	Late rates between seller states and customer states

4. Logic: Big Data Algorithm and Technology Explanation

4.1 Pipeline

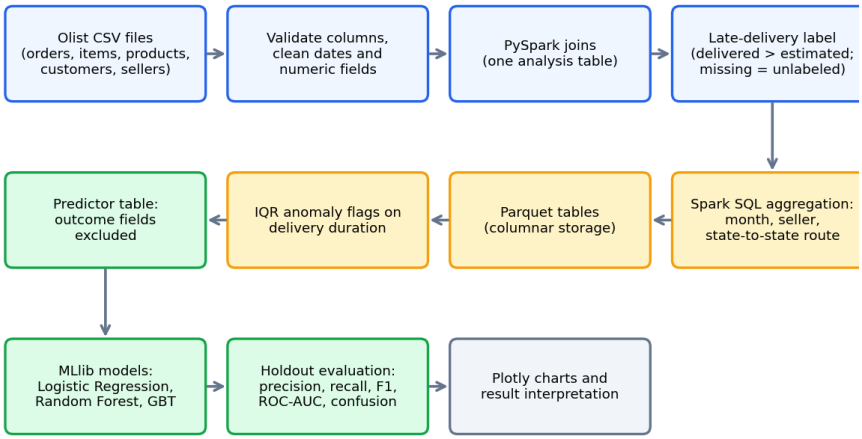


Figure 3. SupplyChainX pipeline from the raw CSV files to model evaluation.

4.2 Apache Spark and PySpark

Apache Spark is a data processing engine that works on DataFrames, which are tables whose operations (filters, joins, aggregations) Spark can split into tasks and run in parallel. PySpark is its Python interface. In SupplyChainX, PySpark reads the CSV files, cleans the date and numeric columns, and performs the joins that build one analysis table.

4.3 Spark SQL aggregation

Once the label exists, the table is registered as a view and queried with Spark SQL. The same pattern, a GROUP BY with the late rate as the average of the label, is used for three views of delivery performance: by month, by seller, and by seller-state-to-customer-state route. This shows when (month) and where (seller, route) delays are more common.

4.4 Parquet storage

The processed tables are saved as Parquet. Parquet stores data by column and keeps the schema, so queries that need only a few columns read less data. This makes it well suited to storing and querying analysis tables.

4.5 Anomaly flags with the IQR rule

Delivery durations that are unusually short or long compared with the overall distribution are flagged with the interquartile range (IQR) rule.

```
IQR = Q3 - Q1
lower fence = Q1 - 1.5 x IQR
upper fence = Q3 + 1.5 x IQR
```

A duration outside the fences is flagged. The flags identify records for review. They do not prove a cause.

4.6 Prediction models (Spark MLlib)

Model	Idea
Logistic Regression	Combines the predictors into a weighted score and turns it into a probability of being late. Simple and linear.
Random Forest	Builds many decision trees on varied samples and features and combines their votes. Can capture non-linear effects.
Gradient-Boosted Trees	Builds trees one after another, each correcting the errors of the earlier ones. Flexible and often strong on tabular data.

The data is split randomly into training and holdout parts. Models are fitted on the training part and compared on the holdout part, and the best model is chosen by holdout F1.

4.7 Evaluation metrics

Late deliveries are less common than on-time ones, so accuracy alone can be misleading. We therefore report precision, recall, F1, ROC-AUC and confusion counts.

```
Precision = TP / (TP + FP)
Recall    = TP / (TP + FN)
F1        = 2 x Precision x Recall / (Precision + Recall)
```

Here a late order is the positive class. Precision is the share of predicted-late orders that were late. Recall is the share of late orders the model found. ROC-AUC measures how well the model ranks late orders above on-time ones across all thresholds.

4.8 Code excerpt (simplified)

The excerpt below is a simplified illustration of the label and aggregation logic. The full code is in the notebook.

```
from pyspark.sql import functions as F

orders = orders.withColumn("is_late",
    F.when(F.col("order_delivered_customer_date").isNull() |
           F.col("order_estimated_delivery_date").isNull(), None)
     .when(F.col("order_delivered_customer_date") >
           F.col("order_estimated_delivery_date"), 1)
     .otherwise(0))

orders.createOrReplaceTempView("orders")
spark.sql("""
  SELECT date_format(order_purchase_timestamp, "yyyy-MM") AS month,
         COUNT(*) AS orders, AVG(is late) AS late rate
  FROM orders WHERE is_late IS NOT NULL
  GROUP BY month ORDER BY month""")
```

5. Result Interpretation and Visualisation

5.1 Model comparison

The three models were compared on the holdout set.

	model	test_rows	accuracy	precision_late	recall_late	f1_late	roc_auc	true_positive	false_positive	true_negative	false_negative
2	Gradient-Boosted Trees	19099	0.733075	0.154197	0.646024	0.248969	0.754128	845	4635	13156	463
1	Random Forest	19099	0.703859	0.135235	0.616208	0.221794	0.718972	806	5154	12637	502
0	Logistic Regression	19099	0.675585	0.125842	0.628440	0.209694	0.701290	822	5710	12081	486

Selected by holdout F1, then ROC-AUC and accuracy: Gradient-Boosted Trees

Best model by holdout F1: [model name], with an F1 score of [value]. Observed late-delivery rate: [value].

5.2 How to read the results

- A model that always predicts "on time" would score high accuracy because late orders are the minority, so we judge the models on F1, precision, recall and ROC-AUC.
- Confusion counts show how many late orders were found (true positives), missed (false negatives) and wrongly flagged (false positives).
- Higher recall finds more late orders, higher precision means fewer false alarms. F1 balances the two.

5.3 Delay patterns

The month, seller and route views show when and where late deliveries are more common. Read them together with order volume, because a high late rate on very

	few orders is less reliable than a high late rate on many orders. Add your main observations from the Plotly charts here. **5.4 Anomalies** The IQR rule flagged unusually short and unusually long delivery durations for review. These flags show records that look unusual compared with the overall distribution. They do not prove why a delivery was fast or slow. **5.5 Limitations** • The data is historical and specific to Brazil. • The test split is random, not chronological, so the results may not reflect how the model would perform on future orders. • The dataset has no warehouse or carrier details. • Spark runs locally in the Colab runtime. This demonstrates Spark processing but is not a multi-machine cluster. **5.6 Conclusion** SupplyChainX shows how big-data tools can combine logistics records, analyse delivery patterns, flag unusual cases and compare delay-risk models. Spark SQL and Plotly make it possible to see when and where deliveries are late, and Spark MLlib allows a fair comparison of three models using metrics suited to a minority positive class.
Conclusion:	We built SupplyChainX, a PySpark project on the public Olist Brazilian E-Commerce dataset (around 100,000 orders, 2016 to 2018). It joins several CSV files, defines late deliveries, compares delivery performance by month, seller and route with Spark SQL, stores processed tables as Parquet, flags unusual delivery durations with the IQR rule, and compares Logistic Regression, Random Forest and Gradient-Boosted Trees in Spark MLlib using precision, recall, F1, ROC-AUC and confusion counts. The results describe historical Brazilian data on a local Spark runtime.
Mini project link	https://drive.google.com/file/d/1YVJ9D3k7MuXXsXvSH-SIma0zUIM9N6_6/view?usp=sharing
References:	[1] Olist. Brazilian E-Commerce Public Dataset by Olist. Kaggle: https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce [2] Apache Spark documentation: https://spark.apache.org/docs/latest/ [3] Spark SQL, DataFrames and Datasets Guide: https://spark.apache.org/docs/latest/sql-programming-guide.html [4] Spark MLlib guide: https://spark.apache.org/docs/latest/ml-guide.html [5] Apache Parquet documentation: https://parquet.apache.org/docs/ [6] Plotly Python documentation: https://plotly.com/python/ [7] Tukey, J. W. (1977). Exploratory Data Analysis. Addison-Wesley.